Source Covers and Barriers to Tape Transposition

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Abstract

We study linear-time transposition of a $d \times d$ matrix with $b(d) = \max(1, \lceil \log_2 d \rceil)$-bit entries on a fixed finite number of tapes. A source certificate must retain every input on which an output can depend and charge the information used to simplify that dependence. We prove a finite correctness bound using the circular gaps in complete raw-source sets, then strengthen it by fractional covers, anchored approximation, and shared raw revelations. For fixed linear runtime and capacity bounds and a fixed bound on the number of disjoint-span classes in the full family system, we obtain a uniform positive correctness deficit in large dimensions for a two-class core plus at most one arbitrary exceptional family and sufficiently small remaining families. No size envelope on the core is needed. With a fixed number q of exceptional families, near-full correctness forces a quantitative lower bound on the tail of the sorted core-family sizes. We also give uniform exclusions under economical source-cover or global-sharing conditions. A native example shows that quadratic pointwise cover growth is possible even for two serial exceptional families. Two complementary results identify the movement cost of a specified coded-caching query interface and the information needed to isolate narrow consecutive column groups causally. Family accounting, clipped-shadow and successful-word variation, protected phases, the exact packed source bank, paid residual recovery, cyclic freezing, interval-cover rounding, the fractional raw-source gap bound, anchored erosion, the full paid-cover theorem, its terminal-orbit corollary and the attached short-distance count are Lean-verified, as are the joint-profile setup, subsequence existence, mass bounds, vanishing tails, the necessary hop-tail inequality, its critical-moment consequence and the uniform one-exception and common-envelope correctness deficits, the finite-skeleton exclusion, the necessary reciprocal-cover and inverse-square probability bounds, the uniform cover criteria, the complete continuation suppliers, and the necessary complete-observable movement bound with its quadratic refinement, and the finite shared-support payment and uniform global-sharing theorem. The envelope criterion needs only finitely many approximate leading-rank bounds; the other numbered results have complete written proofs and remain unformalized. The general transposition question remains open: the sufficient certificate conditions are not proved for arbitrary tape computations.

Contributions. Hans Brende chose the problem and directed the research. GPT-6 Astra (OpenAI) developed the mathematical arguments, Lean formalization and text under his direction. This draft is partially formalized; each result states its proof status.

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1 Introduction

Let $b(d) = \max(1, \lceil \log_2 d \rceil)$ and $L = d^2 b(d)$. Question (T) asks whether one deterministic multitape Turing machine, with a fixed finite number of tapes and a fixed finite alphabet, transposes all such matrices in $O(L)$ time. The input and output are written in row-major order, so the output lists the original columns consecutively. The question is motivated in part by the reductions between transposition and integer multiplication of Harvey and van der Hoeven [8].

An obstruction to routing alone does not settle (T). A tape computation may store nonlinear functions of many input bits and later combine several such records. The first paper of this series isolates that distinction [3]. The depth-reduction program of the fourth and eighth papers cuts a dependency graph by revealing a small transcript, and then asks for a coding bound on the remaining recovery relation [2, 1]. The seventh paper proves half-rate bounds under explicit elimination hypotheses [4]. This paper instead constructs complete source certificates for particular return geometries and proves that their information deficit exceeds the cost of the revelations.

1.1 The argument and its scope

Fix one full computation geometry and a nonempty set $\mathcal{C}$ of inputs correctly transposed on that geometry. Write

$$\Delta = L - \log_2 |\mathcal{C}|.$$

A transcript of at most g bits partitions $\mathcal{C}$ into at most 2^g fibres. If, on every fibre, the complementary coordinates determine a set A of input coordinates, then

$$\Delta \geq |A| - g. \tag{1.1}$$

Our work is to supply A and g from the same actual geometry. Complete source sets retain direct input reads, state and work-tape parents, and unpaid parents entering an output vertex whose record has been revealed. The certificate permits nonlinear computations and correlated correctness sets. It never assigns independent information to different source paths.

The main geometric hypothesis concerns complete return families. A Q_2 core has two classes of pairwise disjoint full return spans; the definition and the strict clipping convention are given in Section 2. An exceptional family is one complete family, not one work tape. In addition to this decomposition we retain fixed runtime and whole-step capacity constants and a fixed bound on the number of disjoint-span classes in the full family system.

1. *A finite source certificate.* A single cyclic ordering of the raw coordinates produces an acyclic recoverable set. The size of that set is controlled by the circular gap areas of complete source banks, with explicit movement, grid and output-stripe errors. Fractional covers and anchored approximation improve its geometric term without changing the recovery bank.

2. *Uniform restricted impossibility.* A Q_2 core plus one arbitrary exceptional family cannot have both $\Delta = o(L)$ and maximum outside-family size $o(L)$. For q fixed exceptions, every limiting sorted hop profile has tail at least a positive multiple of R^{1-q}. Consequently a common size envelope H/i^p with $p > q$ is excluded.

3. *Further sufficient conditions.* Economical fractional source covers, or sufficiently shared complete raw support paid for once across all outputs, give additional uniform correctness-deficit theorems. The quantifiers keep every finite cut parameter fixed before dimension increases. These are criteria on a native class, not assertions that an arbitrary machine satisfies them.

4. *Construction restrictions and limits.* We exhibit a quadratic pointwise cover obstruction, prove a movement bound for a specified query interface even after reordering, and show that isolating fixed groups of $o(\log d)$ consecutive columns causally requires an almost complete helper. Each result states the interface it constrains.

For an oblivious transposer correct on every input, $\Delta = 0$; a positive uniform deficit therefore excludes its geometry. For an adaptive machine, passing to one geometry may already discard much of the input entropy. We do not infer a large common-geometry correctness set without paying that loss. Nor do we prove that the stronger cover criteria exclude a native class outside the union of all earlier exclusions.

1.2 Relation to the earlier papers and other work

The source-bank construction uses actual return families and the owner ordering studied in the eighth paper [1], and the dependency semantics of the fourth [2]. We give the additional packing, residual-cut and raw-source proofs here. The fifth paper develops complete recovery relations and geometric restrictions on their codes [5]. Its distinction between a complete relation and a convenient list of selected paths is essential throughout the present argument. The local coding barriers of the second paper [6] concern different functional interfaces; they do not supply the source covers assumed here.

The fixed-geometry deficit theorems are complementary to the depth reductions of the eighth paper. A depth bound alone leaves a coding question. Under our different geometric hypotheses we complete the deficit calculation. This is not a general improvement of the seventh paper's half-rate theorem or a proof of the unrestricted coding crux.

The movement example uses an established placement delivery array from coded caching [9]; we credit that code and prove only the stated additional query-movement assertion. The causal-group argument uses finite max-flow/min-cut [7] and the first paper's native routing metric. Its additional content is a paid extraction of partial routing from a locally autonomous computation on correlated helper fibres. We make no claim that the elementary covering, entropy or flow principles are new.

1.3 Proof status and organization

Definitions 2.1, 2.2, 2.4, 3.1 and 4.1, Lemmas 2.3, 2.5, 2.6, 2.8, 3.2 to 3.4 and 8.3, Theorems 2.7, 2.9, 3.5, 4.2, 4.6 and 5.2, Corollaries 3.6, 4.3 to 4.5, 4.7 and 4.8, and Propositions 3.7 and 5.1 are formalized in Lean. They reuse the eighth paper's family geometry, shared incidence, interval-cover, interval-graph, finite-orbit, shifted-grid, causal-recovery, circular-gap and stripe-counting lemmas, the second paper's finite-entropy library, the first paper's native counting library, and Mathlib's compactness, summability and bounded-limit results. Their Lean statements are authoritative; the accompanying outlines are guides to those proofs. The other numbered results have complete written proofs and remain unformalized, including global continuation density, metric-sharing criteria and causal-group results. Cited results state their provenance and formal status where used. The index in Appendix B identifies every exact statement, its current status and the verified declarations. The index itself is not a verification certificate.

Section 2 establishes the model, source bank and finite freezing bound. Section 3 supplies the fractional and approximate cover theorem. Section 4 derives the uniform profile consequences; Section 5 treats raw support shared across outputs. Sections 6 to 8 explain the limits and additional construction restrictions. Section 9 identifies the remaining native geometry and computation questions.

2 Complete source banks and paid recovery

This section proves the common finite statement used in the later exclusions. The bank must cover every surviving raw source. A list of selected return endpoints, without the state and unselected work dependencies, does not suffice.

2.1 Fixed geometries and complete return families

Fix integers $d, b \geq 1$, and put $L = d^2 b$, $W = db$ and $\Omega = \{0, \dots, L-1\}$. The finite results allow arbitrary b; the application to (T) takes $b = b(d) = \max(1, \lceil \log_2 d \rceil)$. A *fixed complete geometry* consists of the input-head and work-head walks on $\{0, \dots, T-1\}$, the halting time, and the distinct increasing append times $t_0 < \cdots < t_{L-1}$, where $T \leq cL$. The walks move by at most one cell per step. For the Lean model, extend each walk constantly after $T-1$; this preserves its unit-step property and all hops before T. The output at time t_r demands input coordinate

$$\pi(jW + ib + k) = iW + jb + k \qquad (0 \leq i, j < d,\ 0 \leq k < b). \tag{2.1}$$

The geometry can be an oblivious schedule or one trace shared by a set of inputs. We make no assertion that an arbitrary adaptive transposer has a trace containing nearly all its input information.

The dependency graph has the state edges $t \to t-1$ and every work-head previous-visit edge $u \to p$, where p is the last earlier visit of that head to its cell at time u. A work edge of length one coincides with a state edge. Coincident edges may be identified for reachability, but an edge is retained whenever any physical dependency supplies it. Input reads are sources, not previous-visit edges. The complete value at a time contains its state after the step, its written work symbols and any output bit. Each such value has at most β bits of capacity, for a fixed $\beta > 0$; this includes all data needed by later steps.

A *genie* is a set Γ of time vertices and its transcript G_Γ consists of those complete values. Thus $H(G_\Gamma) \leq \beta|\Gamma|$. A backward path $v_0 \to \cdots \to v_s$ survives the genie if $v_1, \dots, v_s \notin \Gamma$. In particular, its root is retained even when $v_0 \in \Gamma$: we retain its direct read and every incoming dependency whose lower endpoint is unpaid. The full source relation at an output contains every input coordinate read at the end of such a path, including a path of length zero. The usual recursive evaluation of the dependency graph shows that this full source relation and G_Γ determine the output. This is the cone convention of [2, Section 4.1].

Let $\mathcal{C} \subseteq \{0,1\}^\Omega$ be a nonempty set correctly transposed on the fixed geometry, and write

$$\Delta = L - \log_2 |\mathcal{C}|. \tag{2.2}$$

All selections of families, time maps and paid vertices below depend on the fixed geometry, not on the input values within $\mathcal{C}$. We use ordinary base-two Shannon entropy only for finite random variables.

Write $x(t)$ for the input-head position clamped to $[0, L-1]$. This does not increase its total variation. A read outside the input is a known blank; including the bit at the clamped position supplies extra information and is therefore a valid source overcover. For a real-valued path Y and integers $a \leq z$, put

$$\mathrm{Var}(Y; [a, z]) = \sum_{e=a}^{z-1} |Y(e+1) - Y(e)|, \qquad \mathrm{Var}(Y) = \mathrm{Var}(Y; [0, T-1]).$$

In particular, $\mathrm{Var}(x) \leq T - 1$.

Definition 2.1 (Complete families and the two-class condition)**.** Lean: `SourceCovers.actualOwner`, `SourceCovers.ActualFamily`, `SourceCovers.HasSpanColoring`, `SourceCovers.mem_family_edges_iff`, `SourceCovers.constant_extension_isWalk` [`SourceCovers/FamilyAccounting.lean`]

A long hop (p, u) has $u - p \geq 2$. Its side is upper or lower according as the work head stays strictly above or strictly below its endpoint cell between p and u. Its owner is the first time strictly between the endpoints at which its distance from that cell is largest. A *complete actual family* consists of all long hops of one physical head, on one side, with one owner. For a family G, let o_G be that owner, let (p_G, U_G) be its outermost hop, let $I_G = (p_G, U_G]$ be its full half-open span, and let h_G be its hop count. Families are formed from all actual long hops before any selection or payment. They are never redefined by deleting inconvenient hops.

The full family system has an *N-class span partition* if it partitions into N classes whose spans are pairwise disjoint within each class. A selected core is *Q2* if its complete spans admit such a partition into two classes. Empty classes are permitted. The full-system constant N need not be two.

These are the first-maximum families of [1, Sections 2.2 and 3.2]. In that paper a bounded owner-load supplies a bounded number of disjoint-span classes. Here the existence of a fixed N-class partition is an explicit hypothesis; we do not infer it from the number of physical tapes alone.

Definition 2.2 (Admissible return geometry)**.** Lean: `SourceCovers.AdmissibleGeometry`, `SourceCovers.AdmissibleGeometry.inputLength`, `SourceCovers.AdmissibleGeometry.deficit`, `SourceCovers.AdmissibleGeometry.relativeDeficit`, `SourceCovers.AdmissibleGeometry.orbit_count_deficit` [`SourceCovers/AdmissibleGeometry.lean`]

An admissible geometry with parameters $d, b, c, \beta, N, q, a_0$ consists of a fixed complete geometry and nonempty correctness set $\mathcal{C}$ as above, with $T \leq cL$, whole-step capacity at most β, an N-class partition of its full actual family system, an original Q2 core $\mathcal{K}_0$, and a disjoint set $\mathcal{E}$ of at most q other complete families. Every family outside $\mathcal{K}_0 \cup \mathcal{E}$ has at most a_0 hops. The local computation is deterministic; no linearity or product assumption is imposed on $\mathcal{C}$. The word "admissible" below includes all these premises. In a sequence, c, β, N, q are fixed; the geometry, core and exceptional families may vary with dimension.

The formal object supplies the finite record functions and their local dependence on the actual parent relation, using the clamped input path. It packages these fixed-geometry hypotheses; it does not assert that every such tuple is realized by a native Turing machine. In particular, a cover bound, global recovery and a correctness-deficit inequality are not fields of the object. The finite deficit is derived from these data.

Lemma 2.3 (Family accounting)**.** Lean: `SourceCovers.family_owner_inside`, `SourceCovers.family_hops_nested`, `SourceCovers.family_span_hop`, `SourceCovers.family_size_le_owner_arms`, `SourceCovers.two_mul_sum_family_size_le`, `SourceCovers.sum_family_variation_eq`, `SourceCovers.q2_accounting`, `SourceCovers.mem_boundaryLowers`, `SourceCovers.boundary_lowers_le_crossings`, `SourceCovers.boundary_lowers_le_family_sizes`, `SourceCovers.boundary_lowers_le` [`SourceCovers/FamilyAccounting.lean`]

The hops in a family are strictly nested. If its ends are $u_1 < \cdots < u_{h_G} = U_G$, their starts satisfy $p_1 > \cdots > p_{h_G} = p_G$, and every hop contains o_G strictly. The outer span is itself a hop, and

$$h_G \leq \min(o_G - p_G, U_G - o_G) \leq \frac{U_G - p_G}{2}.$$

For any set $\mathcal{K}$ of complete families, let $m_{\mathcal{K}}(e) = |\{G \in \mathcal{K} : p_G \le e < U_G\}|$ be its overlap at the elementary boundary $e \mid e+1$. For every real path Y,

$$2\sum_{G\in\mathcal{K}} h_G \le \sum_{e=0}^{T-2} m_{\mathcal{K}}(e),$$
$$\sum_{G\in\mathcal{K}} \operatorname{Var}(Y;[p_G,U_G]) = \sum_{e=0}^{T-2} m_{\mathcal{K}}(e)|Y(e+1)-Y(e)|. \tag{2.3}$$

In particular, a Q2 core has $m_{\mathcal{K}}(e) \le 2$, and

$$\sum_{G\in\mathcal{K}} h_G \le T-1, \qquad \sum_{G\in\mathcal{K}} \operatorname{Var}(x;[p_G,U_G]) \le 2\operatorname{Var}(x) \le 2(T-1). \tag{2.4}$$

For a set $\mathcal{U}$ of unselected families, let $Z_{\mathcal{U}}(e)$ be the distinct lower endpoints of its work edges and the state edge crossing $e \mid e+1$, and let $c_G(e) = |\{(p,u) \in G : p \le e < u\}|$. At every boundary inside the computation,

$$|Z_{\mathcal{U}}(e)| \le 1 + \sum_{\substack{G\in\mathcal{U}\\ p_G\le e<U_G}} c_G(e) \le 1 + \sum_{\substack{G\in\mathcal{U}\\ p_G\le e<U_G}} h_G. \tag{2.5}$$

Thus, if every unselected family has at most a hops, this count is at most $1 + m_{\mathcal{U}}(e)a \le 1 + Na$. The same conclusion holds under the local bound $m_{\mathcal{U}}(e) \le N$, without supplying a span partition.

Outline of the formal proof. The formalization imports the eighth paper's actual-hop, owner and family definitions and their verified nesting, outer-span and height bounds. Uniqueness of previous visits makes nesting strict. After reflection for a lower family, its hop count is at most the height of the common owner above the outer endpoint cell. The unit-step distance bound limits that height separately by $o_G - p_G$ and $U_G - o_G$, giving the owner-sensitive estimate.

Index an elementary edge by its lower time e. The full span $(p_G, U_G]$ contains its upper time $e+1$ exactly when $p_G \le e < U_G$. Double counting these incidences gives Eq. (2.3): use unit edge weights for span lengths, and $|Y(e+1)-Y(e)|$ for variation. The shared counting identity works for weights in any additive commutative monoid. Families active at one boundary have distinct colors in any disjoint-span coloring. A Q2 core therefore has overlap at most two; the input's unit-step bound gives Eq. (2.4).

For the boundary count, the Lean set contains the state lower endpoint and the union of all crossing-hop lowers in the active families. Its membership theorem proves that this restriction to active families omits no crossing hop. The finite-union and image cardinality bounds give Eq. (2.5). Applying the family-size bound a and overlap bound N gives $1 + Na$. This last counting argument needs neither unit steps nor nesting once the local overlap and size bounds are supplied. □

The source-bank and grid-payment arguments below use the uniform specializations. The exact overlap and crossing-hop profiles retain information that can reduce a geometry's payment estimate; no stronger uniform exclusion is inferred from them here.

2.2 Clipped shadows and exact core orbits

Definition 2.4 (Paid clipping and selected orbits)**.** `Lean`: `SourceCovers.ActualFamily.shadow`, `SourceCovers.ActualFamily.shadow_at_hop`, `SourceCovers.ActualFamily.shadow_exact`, `SourceCovers.shadowOrbit` [`SourceCovers/ClippedShadows.lean`]

Pay the outer lower endpoint p_G of each selected family. Its *clipped shadow* s_G is the identity off $(o_G, U_G]$. On that interval it initially holds o_G; at each hop end u_i it takes the value p_i and holds it

until the next end, with every held value replaced by its maximum with $p_G + 1$. For a family set $\mathcal{S}$, let $\mathcal{O}_{\mathcal{S}}(t)$ be the set of values obtained from t by all finite compositions of its clipped shadows, including the empty composition.

The shadow is antitone on its active interval, and every nonidentity action satisfies

$$p_G < s_G(t) \le o_G < t \le U_G. \tag{2.6}$$

Clipping changes an actual hop value only at the outermost end U_G, whose lower endpoint has been paid. Hence every selected actual hop with an unpaid lower endpoint is represented exactly. We do not gate these maps at other paid vertices. Their orbits may retain extra paths through paid vertices, which is safe for a source overcover and preserves closure under each selected shadow.

For a path Y in a metric space, use the same variation notation with $\operatorname{dist}(Y(e), Y(e+1))$ in place of the absolute difference. This agrees with the earlier definition for real paths.

Lemma 2.5 (Variation of a shadow or successful path)**.** Lean: `SourceCovers.jointShadow_eq_on`, `SourceCovers.jointShadow_variation`, `SourceCovers.shadow_success_cons_iff`, `SourceCovers.shadow_word_structure`, `SourceCovers.shadow_word_variation`, `SourceCovers.variation_comp_le_crossings`, `SourceCovers.exists_sharp_patch` [`SourceCovers/ClippedShadows.lean`]

The joint map of clipped shadows with pairwise disjoint full spans has

$$\operatorname{Var}(Y \circ s) \le 3 \operatorname{Var}(Y)$$

for every metric-valued path Y. For any fixed word of clipped shadows, the set of starting times at which every action is nonidentity is an integer interval, possibly empty. Its endpoint map is monotone in one direction on that interval and no greater than its argument. Extending that endpoint map by the identity outside the success interval also gives variation at most $3 \operatorname{Var}(Y)$ for every Y. The word may be empty and may repeat families.

More generally, for a time map $f : \{0, \dots, n\} \to \{0, \dots, N\}$, let $c_f(z)$ count the indices $e < n$ for which $\min(f(e), f(e+1)) \le z < \max(f(e), f(e+1))$. Then

$$\operatorname{Var}(Y \circ f; [0, n]) \le \sum_{z=0}^{N-1} c_f(z) \operatorname{dist}(Y(z), Y(z+1)). \tag{2.7}$$

The factor three is attained in the general class of regressive interval patches, even by one constant active map.

Outline of the formal proof. The concrete held value is the minimum, over all hops in a complete family, of the hop's start if it has ended and the owner otherwise. It is antitone in time. Strict nesting identifies this minimum with the actual start at each hop end. Taking its maximum with $p_G + 1$ gives exactly the clipped shadow: every hop whose lower endpoint is unpaid is preserved. Its active interval is exactly its nonidentity set.

For the reusable variation argument, sum the triangle inequality along the elementary edges crossed by each jump, then interchange the finite sums. The shared incidence identity gives Eq. (2.7). A map monotone in either direction on one interval and equal to the identity elsewhere has at most one downward crossing of each threshold. Indeed, two downward crossings would force the intervening points into the active interval. An antitone map could not rise between them; for a monotone map, both downward crossings would have to occur at the two joins, contradicting the order of the exterior identity values. Telescoping the threshold's zero–one indicator bounds the total

number of crossings by twice the downward count plus one, hence by three. This argument needs only preservation of the finite time range, not regression of the patch.

For a disjoint packing, a threshold outside a family's full span is unaffected: both the argument and its image stay on the same side. At most one full span straddles a given threshold, so the packed map has the same threshold indicator as that one shadow, or as the identity. The three-crossing bound therefore holds independently of the family count.

Induct on a word. Its success domain is the first action's interval intersected with the preimage of the tail's success domain. Mathlib's order-convex preimage lemmas make this an interval. Composing the monotone restrictions preserves one of the two orientations, and every action weakly decreases time even when inactive. The successful endpoint map therefore has the required interval, monotonicity and range properties. Apply the single-patch bound to its identity extension. Empty success domains and the empty word are included.

For sharpness of the general patch bound, take $f(t) = 1$ on $(2, 4]$ and $f(t) = t$ elsewhere, on times $0, \dots, 5$. With $Y(t) = 0$ for $t < 2$ and $Y(t) = 1$ otherwise, $\mathrm{Var}(Y) = 1$ and $\mathrm{Var}(Y \circ f) = 3$. Lean checks this finite witness directly. No sharpness claim for a particular machine's source bank is inferred from this example. □

This lemma supplies the any-argument-path bound used repeatedly in the exact source bank, without a factor exponential in successful word length. The crossing profile can improve a particular map's estimate; the later uniform constants still use the factor-three specialization. The packed bank is formalized next; recovery through the residual dependencies remains the subsequent task.

Lemma 2.6 (The Q2 overlap forest and protected phases)**.** `Lean`: `SourceCovers.q2_caterpillar`, `SourceCovers.protected_survives`, `SourceCovers.admissiblePhases_card`, `SourceCovers.phase_payment_sum`, `SourceCovers.phase_average_exact`, `SourceCovers.protected_phase_average`, `SourceCovers.nonleaf_mass_le_total`, `SourceCovers.phase_path_length`, `SourceCovers.phase_success_length`, `SourceCovers.phase_shadowOrbit_exact`, `SourceCovers.phase_orbit_chains`, `SourceCovers.phase_orbit_card` [`SourceCovers/ProtectedPhases.lean`]

Join core families whose half-open spans intersect. For a Q2 core this graph is a caterpillar forest: the vertices with at least two neighbors, called the original nonleaves*, form a path within each component where they occur. Let $\mathcal{N}$ be their set. Rank these paths from left to right, starting at one separately in each component. All these choices refer to the core before deletion.*

Fix $M \geq 2$ and a protected set $\mathcal{P}$ of at most $R < M$ core families. Let $\mathcal{A}$ be the residues modulo M not used by the ranks of $\mathcal{P} \cap \mathcal{N}$, and let F be the number of distinct residues they use. For $\rho \in \mathcal{A}$, delete only the original nonleaves of rank congruent to ρ, paying every actual lower endpoint of their hops; call this union $\Gamma_{\mathrm{rank}}(\rho)$. Every protected family survives. Put $\tau_{\mathcal{N}} = \sum_{G \in \mathcal{N} \setminus \mathcal{P}} h_G$. Then

$$|\mathcal{A}| = M - F \geq M - R, \qquad \frac{1}{|\mathcal{A}|} \sum_{\rho \in \mathcal{A}} |\Gamma_{\mathrm{rank}}(\rho)| \leq \frac{\tau_{\mathcal{N}}}{|\mathcal{A}|} \leq \frac{\tau_{\mathcal{N}}}{M - R}. \tag{2.8}$$

More precisely, the sum of paid-union sizes is at most $\sum_{G \in \mathcal{N} \setminus \mathcal{P}} |L_G|$, where L_G is the set of actual hop lowers of G. The average bound with denominator $M - F$ only requires $F < M$, regardless of $|\mathcal{P}|$.

In every phase, each surviving simple overlap path has at most $M + 1$ vertices; in particular each successful core word has at most $M + 1$ actions. The complete core orbit is the union of at most two deterministic chains from the root and has at most $\ell = 2M + 3$ time values. Original leaves and components of one or two vertices need not be deleted.

Outline of the formal proof. The overlap graph has no triangle: three pairwise intersecting intervals have a common point, but one Q2 class would contain two of them. It has no longer cycle either. On a cycle choose a vertex with largest left endpoint. Its two cycle neighbors both extend strictly past that endpoint, hence overlap each other, giving a triangle. Thus the graph is a forest. Lean reuses Mathlib's cycle rotation and path operations.

An interval contained in another can have no further neighbor without creating a triangle. Consequently nonleaf intervals have distinct left endpoints and strictly increasing right endpoints in that order. An overlap edge between nonleaves cannot skip a third nonleaf in the order: that third interval would meet both endpoints and create a triangle. Lean defines the rank as one plus the number of earlier nonleaves in the same connected component, and proves that adjacent ranks differ by one and distinct nonleaves have distinct ranks. Every internal vertex of a simple path is a nonleaf, so removing original leaves preserves connectivity between nonleaves. These facts give the connected nonleaf path and its maximum degree two.

A finite unit-step integer chain avoiding one residue modulo M visits at most $M-1$ distinct integers. Lean proves this using the first forbidden integer at or above the chain's minimum; a unit step cannot cross that integer without visiting it. Apply this to the internal vertices of a surviving simple path. Their ranks are distinct, so including the two endpoints gives at most $M+1$ vertices. This argument needs no deletion of leaves and also covers components of one or two vertices.

By (2.6), consecutive successful actions have overlapping spans and strictly decreasing owners. Their names form a simple path, to which the preceding bound applies. After an action from one Q2 class, its image lies in the same family's span at or before its owner. That family is now inactive and no other family of its class can act there. The only possible next action is therefore the unique active family of the other class. At the root there are at most two choices of first action; thereafter continuation is deterministic. Including all intermediate stopping points gives at most $1+2(M+1)=2M+3$ values.

The shared finite-orbit proof erases inactive actions from an arbitrary word without changing its endpoint. It identifies the full word orbit exactly with the values reachable in at most $M+1$ nonidentity steps. This includes the empty word, intermediate stopping points and any collisions between the two chains.

Only protected nonleaves forbid residues. Each unprotected nonleaf is deleted in at most one admissible phase. Mathlib's finite-union bound charges its actual lower support, whose size is at most its hop count. Interchanging the finite sums gives the payment inequality, and counting distinct forbidden residues gives $|\mathcal{A}| = M - F$. □

This supplies the finite core-orbit bound needed for the exact source bank. Compared with deleting leaves together with their neighbors, it retains the same orbit bound while paying only for original nonleaves. The earlier estimate using $\tau = \sum_{G\in\mathcal{K}\setminus\mathcal{P}} h_G$ still follows from $\tau_{\mathcal{N}} \le \tau$. The next theorem assembles these bounds into a uniform packed bank.

Theorem 2.7 (Theorem S: an exact source bank)**.** Lean: `SourceCovers.exact_source_bank`, `SourceCovers.exactSourceBank_length`, `SourceCovers.exactSourceBank_bounded`, `SourceCovers.exactSourceBank_exact`, `SourceCovers.exactSourceBank_cover`, `SourceCovers.exactSourceBank_raw_root`, `SourceCovers.exactSourceBank_raw_values_card`, `SourceCovers.exactSourceBank_raw_selected_edge`, `SourceCovers.exactSourceBank_raw_variation` [`SourceCovers/ExactSourceBank.lean`]

Select a Q2 core of at most J complete families, at most q further complete exceptional families, and a protected subcore of size at most R. Let $M \ge \max(2, R+1)$ and use any admissible phase of Lemma 2.6. Pay the rank lowers and the outer lower endpoints of all selected core and exceptional

families. Put

$$\begin{aligned} \ell &= 2M+3, \qquad m_0 = 4+2M(M-1), \qquad C_0 = 16+24M(M-1), \\ m_q &= m_0(1+m_0)^q, \qquad C_q = C_0(1+3C_0)^q. \end{aligned} \tag{2.9}$$

There is a list $\mathcal{B}$ of at most m_q geometry-fixed total time maps such that, simultaneously for all time vertices t,

1. *$\{a(t) : a \in \mathcal{B}\}$ is exactly the complete orbit of the retained core and all selected exceptional shadows;*
2. *for every real-valued path Y, $\sum_{a\in\mathcal{B}} \operatorname{Var}(Y \circ a) \le C_q \operatorname{Var}(Y)$;*
3. *the time-value set is a union of at most $(\ell+1)^q$ complete core orbits, each a union of the two chains in Lemma 2.6.*

Consequently the raw bank $Y_a = x \circ a$ contains the root read, has total variation at most $C_q(T-1)$, and is set-closed across every selected actual edge whose lower endpoint is unpaid. Its pointwise cardinality is at most $\ell(\ell+1)^q$. For fixed M and q, the constants do not depend on J or on the sizes or scales of the original families.

Outline of the formal proof. Keep the original nonleaves as spine vertices. In each two-vertex component, add its color-zero vertex to the spine; leave isolated vertices in the leaf pack. The remaining leaves have pairwise disjoint spans, so their joint shadow s has variation constant three. Every internal vertex of a successful word is an original nonleaf. Thus packing leaves occur only at the two ends of such a word.

Label a surviving nonleaf by its original rank modulo M, excluding the deleted residue. These $H = M-1$ labels are injective within each surviving spine component: along a path, ranks make unit steps and avoid the deleted residue, so they stay in an interval shorter than one period. Equal residues therefore mean equal ranks and equal vertices. The extra spine vertices from two-vertex components are isolated in the spine and may use any of the H labels. This uses $M \ge 2$.

For each unordered pair of labels, with repetition allowed, pack one map across all surviving spine components. In a component, choose the nonempty successful word with those endpoint labels, if one exists. The forest has a unique simple path between two vertices, and strictly decreasing owners fix its orientation. Hence that word is unique. Use its endpoint map on its success interval and identity elsewhere. If no word exists, use identity throughout that component.

The union of the spans in a connected component is its interval hull, and distinct component hulls are disjoint. Each chosen word remains in its component hull. The shared disjoint-patch theorem therefore bounds each packed map's variation by three, including both joins to identity. Add an identity slot. Erasing inactive actions from any spine word shows that these slots give exactly its complete orbit, simultaneously at every root. Their number and summed variation coefficient are

$$m_S = 1 + \binom{M}{2}, \qquad C_S = 1 + 3\binom{M}{2}.$$

All choices precede the root and the argument path; a slot's representing word is allowed to vary between components.

For each spine slot a, use

$$a, \qquad a \circ s, \qquad s \circ a, \qquad s \circ a \circ s.$$

The two endpoint leaves and the spine middle of every successful core word are represented by these four choices. Conversely each value follows a core word at that argument. This proves exactness,

rather than merely coverage. The any-path variation bound composes: the four contributions are at most $C_S, 3C_S, 3C_S, 9C_S$. Thus there are $4m_S = m_0$ slots with total coefficient $16C_S = C_0$. The pointwise bound remains ℓ, by Lemma 2.6.

Sort the exceptional families by decreasing owner, retaining a fixed tie order. Successful words have strictly decreasing owners, so each such word uses an ordered exceptional subsequence, with core words between its members. Let B be the core bank and D the bank for the remaining exception list. On inserting its first shadow f, form

$$D \;{+}{+}\; \{b \circ f \circ a : b \in D,\ a \in B\},$$

where $+\!+$ denotes list concatenation. Exactness follows by splitting a successful word at f, or omitting f if it is absent. Every constructed value follows a selected word, even though a packed core slot need not represent one fixed word globally. Slot count and summed variation obey

$$m_{r+1} = (1 + m_0)m_r, \qquad C_{r+1} = (1 + 3C_0)C_r.$$

These recurrences give (2.9) without separately formalizing the binomial sums over subsequences.

For the complete-core-orbit cover, retain the starting vertices of each final core continuation. Each inserted exception creates at most ℓ new choices for each old continuation pattern, in addition to the pattern omitting it. Induction gives at most $(\ell + 1)^q$ terminal core roots. The equality of the covered set with the enlarged orbit proves its pointwise cardinality bound.

The empty word gives the root read. Prefixing a selected shadow gives orbit set-closure. For an original core family removed by the phase, every actual lower endpoint lies in the rank payment; it therefore has no unpaid actual edge. For every remaining selected family, strict clipping represents the actual hop exactly unless its outer lower was paid. This proves closure across all originally selected unpaid edges, and taking images under x preserves it. Finally, the unit-step bound $\mathrm{Var}(x) \le T - 1$ gives the raw variation estimate. Other paid vertices do not truncate the orbit. □

The formal variation bound holds for every pseudometric-valued path and every finite horizon, with the same constants. The geometric bank needs no unit-step assumption on the work walks; that assumption is used when identifying shadows with actual hops. It also needs no numerical bound on the number of core families. In the generic exceptional-list argument, repeated entries and overlap with the core are allowed; the displayed bounds count list entries.

Theorem S supplies a representation of complete selected dependence whose variation cost, for fixed M and q, is independent of the number of selected core families. The next lemma pays for the remaining actual dependencies. Without this step, Theorem S alone is not a transposition certificate.

2.3 Residual cuts and complete functional recovery

For an integer width $w \ge 1$ and shift $\theta \in \{0, \dots, w - 1\}$, define the linear, endpoint-clipped block

$$B_\theta(y) = \{z \in \Omega : \lfloor (z + \theta)/w \rfloor = \lfloor (y + \theta)/w \rfloor\}. \tag{2.10}$$

Thus $|z - y| \le w - 1$ within a block, including a partial block at either end of the input.

Lemma 2.8 (Paid residual-source recovery)**.** `Lean`: `SourceCovers.paid_residual_recovery` [`SourceCovers/ResidualRecovery.lean`]

Let Γ_0 be an initial paid set, and let $Y_1, \dots, Y_m$ be geometry-fixed maps from the timeline to Ω. Their value set $A(t)$ contains $x(t)$. Partition all dependency edges into selected and residual edges.

Suppose $A(u) \subseteq A(v)$ for every selected edge $v \to u$ with $u \notin \Gamma_0$. For an unpaid lower endpoint u, let $m_R(u)$ be its latest residual successor, or u if none. Write $z_{u,i} = \operatorname{osc}(Y_i; [u, m_R(u)])$, where oscillation is maximum minus minimum, and let $\mathcal{U}$ be the set of initially unpaid lowers with an actual residual successor. Define

$$V = \sum_i \operatorname{Var}(Y_i), \qquad \mathcal{R} = \sum_{u \notin \Gamma_0} \sum_i z_{u,i}, \qquad \mathcal{R}_w = \sum_{u \in \mathcal{U}} \sum_i \min\{w, z_{u,i}\}.$$

At the elementary boundary $e \mid e+1$, put

$$h_e = |\{u \in \mathcal{U} : u \le e < m_R(u)\}|, \qquad D_e = \sum_i |Y_i(e+1) - Y_i(e)|.$$

If $h_e \le H$ at every boundary for a real $H \ge 0$, then, for every integer $w \ge 1$, there are sets $\Gamma_\theta \subseteq \mathcal{U}$ satisfying

$$\begin{aligned} \mathbb{E}_\theta |\Gamma_\theta| &\le \frac{\mathcal{R}_w}{w} \le \frac{\mathcal{R}}{w}, \\ \frac{\mathcal{R}}{w} &\le \frac{1}{w} \sum_{e=0}^{T-2} h_e D_e \le \frac{HV}{w}. \end{aligned} \tag{2.11}$$

For each shift, after paying $\Gamma = \Gamma_0 \cup \Gamma_\theta$, every output at time t_r is determined on $\mathcal{C}$ by the same transcript G_Γ and the bits in

$$\bigcup_{i=1}^{m} B_\theta(Y_i(t_r)). \tag{2.12}$$

No bound on full path depth, linearity or independence on a genie fibre is needed.

Outline of the formal proof. For each unpaid residual lower u, pay u if, for some i, the label $\lfloor (Y_i(t) + \theta)/w \rfloor$ is not constant on $[u, m_R(u)]$. This is an actual union of lower endpoints, so a lower used by several dependencies is paid once. If the integer range of one slot on an interval has oscillation z, exactly $\min(w, z)$ of the w shifts cut that range. The formal count reuses the bijection of residues over one period. The union bound over slots and lowers gives the capped cost $\mathcal{R}_w$. An unpaid vertex with no residual successor has a singleton lifetime and zero oscillation, so omitting these vertices changes neither cost. Oscillation is at most variation on the interval. The shared finite incidence identity sums those interval costs with exact multiplicity h_e. A lifetime crosses $e \mid e+1$ exactly when its lower has an actual residual successor after e. This gives the profile bound before using $h_e \le H$ and $\sum_e D_e = V$.

A surviving residual edge preserves every slot label. A surviving selected edge preserves inclusion of the sets of labels by the hypothesis. Along any surviving backward path, induct from its last vertex to its root: the label of its direct input read belongs to the root's label set, because $x(t) \in A(t)$ at the last vertex. Therefore every full surviving raw source belongs to (2.12). This treats paths of arbitrary length and all alternations of state and physical work edges.

Finally compare two inputs in $\mathcal{C}$ with the same transcript and the same bits in the displayed union. Strong induction on time gives equality of their records: a paid parent's equality comes from the transcript, and an unpaid parent's equality comes from the induction hypothesis. Their direct reads agree by the cone containment, so the local deterministic rule gives equality of the current record. Mathlib's factorization theorem then supplies a decoder of the whole record, hence of its output component. Initial values and reads outside the input are constants. The same reasoning starts at a revealed output root and retains its direct read and all unpaid parents; revealing the root was never used to delete them. □

The capped cost and the actual boundary profile can improve the original HV/w estimate without changing the recovery sets. The formal theorem uses $n = T - 1$ as the last time and proves recovery of the whole step record in an arbitrary causal dependency graph. Its local premise is that the direct read and all parent records determine the current record on $\mathcal{C}$; it does not assume global recovery. Fixed initial values and outside-input blanks are constants in this local rule.

For Theorem S, take the surviving selected family edges as selected, retain all other state and physical work edges as residual, and disregard only edges already blocked at their lower endpoints. If every family outside the selected core and the selected exceptional set has at most a hops, Lemma 2.3 gives $H \le 1 + Na$. In each admissible phase there is a shift with

$$|\Gamma| \le J + q + |\Gamma_{\mathrm{rank}}(\rho)| + \frac{(1 + Na)C_q(T - 1)}{w}. \tag{2.13}$$

In particular a phase attaining the average in (2.8) permits replacing its rank payment by $\tau_{\mathcal{N}}/(M-R)$, or the older upper bound $\tau/(M - R)$. More generally, one may average a complete signed reward-minus-payment expression over the admissible phases and their raw-grid shifts. A reward chosen in one phase or shift cannot be paired with a payment from another. Deleted selected families are blocked by their paid lowers; they are not silently reclassified as small residual families.

2.4 Cyclic freezing of actual transpose demands

View Ω as the integer points on a circle of circumference L. For a nonempty subset S, list the circular distances between consecutive distinct points as $g_1, \dots, g_v$; their sum is L. A singleton has one gap of length L. Define its point-gap area by

$$\mathcal{A}(S) = \frac{1}{2L^2} \sum_{h=1}^{v} g_h^2. \tag{2.14}$$

The blocks in (2.10) remain linear blocks. The circle is used only for an overcover and a common ordering of raw coordinates.

Theorem 2.9 (Theorem F: cyclic freezing)**.** Lean: `SourceCovers.cyclic_freezing`, `SourceCovers.cyclic_freezing_oscillation`, `SourceCovers.cyclic_freezing_genie`, `SourceCovers.cyclic_freezing_genie_raw` [`SourceCovers/CyclicFreezing.lean`]

Let $m \ge 1$ and let $Y_i : \{0, \dots, T-1\} \to \Omega$, $1 \le i \le m$, be geometry-fixed bank maps. Let Z be a deterministic helper determined by the input, with $H(Z) \le g$ for the uniform law on $\mathcal{C}$. Suppose every output is determined on $\mathcal{C}$ by Z and the raw block union in (2.12), at one width w and shift θ. Put $V = \sum_i \mathrm{Var}(Y_i)$. For output column j, let $s_j = t_{jW}$ be its first append time and set $S_j = \{Y_i(s_j) : 1 \le i \le m\}$ and $A_j = \mathcal{A}(S_j)$. Then

$$\Delta \ge W \sum_{j=0}^{d-1} A_j - \frac{2V}{d} - 2(w-1) - \frac{W - b + 1}{2} - g. \tag{2.15}$$

More precisely, write $e_j = t_{(j+1)W-1}$ and

$$\rho_j = \max_{1 \le i \le m} \left(\max_{s_j \le t \le e_j} Y_i(t) - \min_{s_j \le t \le e_j} Y_i(t) \right).$$

The bound holds with V replaced by $\sum_j \rho_j \le V$. In particular it holds with $g = \beta|\Gamma|$ for an actual whole-step genie. If u additional raw bits at geometry-fixed coordinates are revealed jointly with that genie, it holds with $g = \beta|\Gamma| + u$. The bound permits arbitrary nonlinear recovery and a correlated correctness set. Its area term uses the actual transpose demand stripes, not an assumption that demands and sources are independent.

Outline of the formal proof. Freeze the bank at each column's first append time and use the fixed circular overcover

$$T_j = \{z \in \Omega : \mathrm{dist}_{\mathrm{circ}}(z, S_j) \le r_j\}, \qquad r_j = w - 1 + \rho_j.$$

Every coordinate of a source block differs from its current anchor by at most $w-1$, and that anchor differs from its frozen value by at most ρ_j. This covers entire linear blocks. The column time intervals have disjoint elementary edges, so $\sum_j \rho_j \le V$ by the finite-path variation bound. The formalization reuses the first paper's transpose-position formula to identify each output's actual column window.

Choose a uniformly random circular cut and an independent uniformly random orientation. They define one total order on all raw coordinates, shared by every column. Select a demanded coordinate z of column j when it precedes every coordinate in T_j. Strict precedence rejects $z \in T_j$ automatically. Every recovery edge between selected coordinates points forward in this same order.

A consecutive frozen-anchor gap of length g contains an eroded arc of $a = (g - 1 - 2r_j)_+$ integer points. These arcs are pairwise disjoint and avoid T_j. For the v-th point of a nonempty arc, v cuts in one orientation and $a + 1 - v$ cuts in the reverse orientation safely place that point before all of T_j. Thus its selection probability is at least $(a+1)/(2L)$. Only this injection of safe choices is needed; we need not enumerate every possible selection or every complementary run. The sum of the guaranteed weights over nonempty eroded arcs is at most $1/2$, since $a + 1 \le g$ for each such arc and the original gaps sum to L.

Consequently the mean selection probability over all L raw coordinates is at least

$$A_{j,r_j} = \frac{1}{2L^2} \sum_g (g - 1 - 2r_j)_+ ((g - 1 - 2r_j)_+ + 1).$$

For integers $g \ge 1, r \ge 0$,

$$g^2 - (g - 1 - 2r)_+ ((g - 1 - 2r)_+ + 1) \le (4r + 1)g.$$

Expansion proves the case $g \ge 2r + 1$; otherwise the left side is $g^2 \le (4r + 1)g$. Summing gives

$$A_{j,r_j} \ge A_j - \frac{2r_j + 1/2}{L}. \tag{2.16}$$

There is no factor for the number of slots or distinct frozen points.

The actual demanded coordinates in column j form the stripe

$$D_j = \{iW + jb + k : 0 \le i < d,\ 0 \le k < b\}.$$

For every circular integer interval I,

$$\left| |(|D_j \cap I) - \frac{|I|}{d} \right| \le b(1 - 1/d). \tag{2.17}$$

Extend the stripe indicator periodically with period W. Its centered prefix count, obtained by subtracting n/d from the count on $[0, n)$, has oscillation $b(1 - 1/d)$: it rises with slope $1 - 1/d$ on the occupied block and falls with slope $-1/d$ on its complement. Interval discrepancy is a difference of two prefix values. Since L is a multiple of W, this also handles intervals across the circular seam.

Apply (2.17) to the disjoint eroded arcs, weighted by their guaranteed selection probabilities. The weights sum to at most $1/2$, so the expected selected demand count in column j is at least

$$WA_{j,r_j} - \frac{b(1 - 1/d)}{2} \ge WA_j - \frac{2r_j + 1/2}{d} - \frac{b(1 - 1/d)}{2}.$$

Sum over the actual column stripes, which partition Ω, and use $\sum_j r_j = d(w-1) + \sum_j \rho_j$. This gives the geometric part of (2.15), with V replaced by $\sum_j \rho_j$. Retaining the factor $1 - 1/d$ improves the former additive error $(W+1)/2$ by $b/2$; the earlier bound follows immediately.

For any realized common order, let A be its selected coordinates. Given Z and the raw complement $\Omega \setminus A$, recover selected coordinates in reverse order. Reusing the second paper's finite-entropy library, for X uniform on $\mathcal{C}$ we obtain

$$\log_2|\mathcal{C}| = H(X) \le H(Z) + H(X_{\Omega\setminus A}) \le g + L - |A|.$$

Average over the common cut and orientation. The same helper is used for every order, and no support-cardinality bound is inferred from its entropy. The genie and joint-revelation special cases use subadditivity and the record alphabet's real-valued bit capacity. No independence assumption is needed. □

Theorem F charges the same genie that supplies the complete bank. Own-source demands are rejected in its acyclic selection, so no separate own-block collision loss is added. If M, q, c are fixed, Theorem S gives $V = O(L)$, and the variation and stripe errors are $o(L)$ as $d \to \infty$. The substantive remaining question is whether the point-gap area exceeds the rank and residual payments; the following sections prove this under explicit source-cover hypotheses.

3 Fractional source covers and a paid deficit

We now improve the geometric term in Theorem 2.9. The recovery bank remains the complete bank of Theorem 2.7. The covers introduced here are witnesses used to estimate its raw source sets; they are not additional records available to the machine.

3.1 Parameters and complete core covers

Fix an admissible geometry, a selected Q_2 subcore of at most J families, and at most q complete exceptional families. Choose integers $K, R \ge 0$, $J \ge \max(K, R)$ and $M \ge \max(2, R+1)$, distinguish at most K selected core families, protect at most R selected core families, and choose a legal rank phase ρ. All distinguished families are among those selected before the phase; they need not be protected. Write

$$\ell = 2M + 3, \qquad C_0 = 16 + 24M(M-1), \qquad C_q = C_0(1 + 3C_0)^q. \tag{3.1}$$

Let $A_\rho(u)$ be the complete orbit of the retained core at time u, and $O_\rho(t)$ the complete orbit with the selected exceptions included. The raw source set is $S_\rho(t) = x(O_\rho(t))$. Put

$$\alpha L = \sum_{G \text{ selected core, not distinguished}} \mathrm{Var}(x; [p_G, U_G]), \qquad \tau = \sum_{G \text{ selected core, not protected}} |G|. \tag{3.2}$$

These sums use original full spans and hop counts, before rank deletion; in particular they do not depend on ρ.

Definition 3.1 (Fractional complete-core cover)**.** Lean: `SourceCovers.coreRawSets`, `SourceCovers.fractionalCoreCover`, `SourceCovers.fractionalCoreCover_exists_optimal`, `SourceCovers.one_le_fractionalCoreCover_toReal`, `SourceCovers.coreRawSets_phase_shadowOrbit_eq`, `SourceCovers.fractionalCoreCover_ne_top_iff_subset`,

`SourceCovers.fractionalCoreCover_exactSourceBank_ne_top`
[`SourceCovers/CoreFractionalCover.lean`]

For a nonempty finite set $S_0 \subseteq \{0, \dots, L-1\}$, define $\kappa_\rho(S_0)$ as the minimum of

$$\sum_{u=0}^{T-1} \lambda_u \quad \text{subject to} \quad \lambda_u \geq 0, \qquad \sum_{u:\, a \in x(A_\rho(u))} \lambda_u \geq 1 \quad (a \in S_0).$$

If no cover exists, set $\kappa_\rho(S_0) = +\infty$. Covering orbits may be rooted at any computation time; their raw sets may contain points outside S_0. A finite cover has an optimum, since weights may be truncated at one and the resulting feasible set is compact. It has weight at least one. In fact a finite cover exists exactly when $S_0 \subseteq \{x(u) : 0 \leq u < T\}$: every complete core orbit contains its root and only earlier times. Thus unit weights at all computation times suffice whenever the condition holds, including for $S_0 = S_\rho(t)$ at every computation time t.

Lemma 3.2 (Fractional interval covers)**.** Lean: `SourceCovers.exists_cover_packing`, `SourceCovers.exists_subcover_le_weight` [`SourceCovers/IntervalCover.lean`]

Let S be a finite subset of a linearly ordered set, and let $(A_i)_{i \in I}$ be a finite family of order-convex sets covering S. Order-convex means that each set contains every point between two of its points. There exist $C \subseteq I$ and $P \subseteq S$ such that

$$S \subseteq \bigcup_{i \in C} A_i, \qquad |C| = |P|, \qquad |P \cap A_i| \leq 1 \quad (i \in I).$$

Consequently, nonnegative weights w_i satisfying $\sum_{i:\, x \in A_i} w_i \geq 1$ for every $x \in S$ admit an integral subcover of cardinality at most $\sum_{i \in I} w_i$. The target S may be empty. In particular, this applies to closed, open or half-open real intervals without a boundedness assumption.

Outline of the formal proof. Induct on the finite target. For a nonempty target let x be its least point, and let y be the greatest target point sharing an A_i with x. Choose such a set A_j containing x, y. Order convexity makes A_j cover all target points through y. No set containing x meets the residual target above y, by the choice of y. Apply induction there after removing j from the index family, then add j to the subcover and x to the packing. Both cardinalities increase by one, and every original set still contains at most one packing point. The empty target uses two empty sets.

Fractional coverage implies ordinary coverage. Sum its inequalities over the packing and interchange the finite sums. Each set's weight occurs at most once, so $|C| = |P| \leq \sum_i w_i$. The proof uses only finite extrema, order convexity and finite sums; the shared library supplies the declarations, which this paper re-exports. □

This rounding step preserves the total fractional weight in the raw-source gap bound below. It does not alter the complete recovery bank.

Lemma 3.3 (A fractional raw-source gap bound)**.** Lean: `SourceCovers.fractional_phase_gap`, `SourceCovers.fractional_core_gap`, `SourceCovers.fractional_core_gap_of_union_variation`, `SourceCovers.fractional_core_gap_of_weights` [`SourceCovers/FractionalGap.lean`]

With the parameters above, any nonempty S_0 with finite $\kappa_\rho(S_0)$ satisfies

$$\mathcal{A}(S_0) \geq \frac{(1-\alpha)_+^2}{2(2K+1)\kappa_\rho(S_0)}, \qquad (z)_+ = \max(z, 0). \tag{3.3}$$

Here $\mathcal{A}(S) = \sum_g |g|^2/(2L^2)$, with the sum over the circular gaps of the distinct points in S; a singleton has one gap of length L.

A sharper form retains actual component counts. Join consecutive computation times across the elementary edges in the union B of the original nondistinguished full spans. Let $m_\rho(u)$ count the resulting time components met by $A_\rho(u)$ and put $V_B = \sum_{e\in B} |x(e+1) - x(e)|$. For any feasible cover weights,

$$\mathcal{A}(S_0) \geq \frac{(1 - V_B/L)_+^2}{2\sum_{u=0}^{T-1} m_\rho(u)\lambda_u}. \tag{3.4}$$

Here $V_B \leq \alpha L$ and $m_\rho(u) \leq 2K + 1$. The same conclusions hold for any retained subset of the selected Q_2 core. No rank phase or unit-speed assumption on x is needed for this estimate.

Outline of the formal proof. Label a time t by the number of elementary edges before t that are outside B. Equal labels define contiguous time components, including isolated times. Their internal edge sets partition B within the computation horizon. The interval between the minimum and maximum raw coordinates on each component has length at most its internal variation. Thus the sum of all component interval lengths is at most V_B, with every elementary edge charged once.

A nondistinguished shadow stays in one component, since every edge between its endpoints belongs to its original full span. A complete core orbit has at most two initial successors and thereafter deterministic continuation. Along either branch, a label change uses a distinguished family whose owner precedes the current time. The new time is at most that owner, so this family cannot pay for another change on the branch. Counting the shared root label once gives $m_\rho(u) \leq 1 + 2K$. The formal proof keeps the smaller count of retained distinguished owners preceding the root before relaxing it to K.

Give each component interval the sum of the weights of the covering orbits that meet it. These weights cover S_0 and have total $\sum_u m_\rho(u)\lambda_u$. By Lemma 3.2, they admit an integral subcover with at most that many intervals. Their total length is at most V_B.

Sort the distinct target points and append the first point plus L. Call a consecutive gap exposed if no chosen interval contains both endpoints. One interval can contain the left endpoint of at most one exposed gap: if it contained two, it would contain both endpoints of the earlier gap. Since every left endpoint is covered, the exposed gap count is at most the subcover size. The other gaps have disjoint elementary edge sets inside the covering intervals, so their total length is at most V_B. The exposed gaps therefore have total length at least $(L - V_B)_+$. Cauchy–Schwarz, followed by division by $2L^2$, gives Eq. (3.4). The unrolled last gap handles the circle seam and singleton target without an extra component.

Finally take an attained minimum cover, use $V_B \leq \alpha L$ and $m_\rho(u) \leq 2K+1$, and obtain Eq. (3.3). All extrema, gap lengths and component incidences in this argument are constructed from the actual time path and the complete core orbits. □

This supplies a verified geometric credit for the paid-deficit argument. The shared-root count improves the former factor $4(K + 1)$ to $2(2K + 1)$; the quantitative consequences below use the improved factor. It does not supply economical covers for arbitrary native geometries.

3.2 Anchored approximation

We represent finite circular geometry by a common subdivision containing all points and arc endpoints. Its elementary edges have real lengths $w_e \geq 0$ summing to $L > 0$. A closed arc contains consecutive edges and both endpoint vertices; its union length is the sum of the lengths of the edges in that union. Thus an edge covered several times is counted once. Subdivision indices are labels, not physical integer coordinates. The formalization identifies each weighted gap with its real endpoint displacement, including the gap across the chosen seam. Taking all edge lengths equal to one recovers the raw-coordinate definition used above. Write $\psi(e) = e - e^2/2$.

Lemma 3.4 (Anchored erosion of circular gaps)**.** Lean: `SourceCovers.penalty`, `SourceCovers.area_lower_bound`, `SourceCovers.area_lower_bound_linear`, `SourceCovers.anchored_displacement_area_lower_bound`, `SourceCovers.anchored_point_area_lower_bound` [`SourceCovers/AnchoredErosion.lean`]

Let S_0, S be nonempty finite point sets on the circle of circumference L, represented in a common subdivision as above. Suppose that $S \subseteq S_0 \cup \bigcup_{i=1}^{m} I_i$, where the I_i are closed circular arcs, each containing a point of S_0. If their union has length at most E, set $e = \min\{E/L, 1\}$. Then

$$\mathcal{A}(S) \geq \mathcal{A}(S_0) - \psi(e) \geq \mathcal{A}(S_0) - E/L. \tag{3.5}$$

The statement permits points of S_0 outside S, overlapping arcs, singleton arcs and full-circle arcs. For $0 \leq E \leq L$, the loss bound is $E/L - E^2/(2L^2)$.

Outline of the formal proof. Let U be the union of covered elementary edges and let $C = \sum_{a \in U} w_a$. Label each edge by the gap containing it, first for S_0 and then for S. Two uncovered edges in the same S_0 gap must share an S gap. Indeed, any intervening S point outside S_0 lies in a covering arc. That arc also contains an S_0 point outside the intervening interval, so it must cross one of the two supposedly uncovered edges. This argument uses actual arc membership and does not assume $S_0 \subseteq S$.

For either gap labeling c, expand the squared-gap sum as $\sum_{a,b:\, c(a)=c(b)} w_a w_b$. The preceding implication preserves every old pair whose two edges are uncovered. Consequently the decrease in the squared-gap sum is at most the mass of all ordered pairs touching U, namely

$$L^2 - (L - C)^2 = 2LC - C^2.$$

Divide by $2L^2$ and use $0 \leq C/L \leq \min\{E/L, 1\}$ together with the monotonicity of ψ on $[0, 1]$. Finally $\psi(e) \leq e \leq E/L$ gives the original linear bound directly. The shared proof uses finite partition sums and the previously verified actual circular gaps; the real-coordinate adapter follows by telescoping edge lengths. □

This verifies approximation by anchored arcs while retaining the complete source set in recovery. The quadratic saving increases the available finite geometric credit; it does not establish economical approximations for arbitrary tape geometries.

The anchoring premise matters: adding the isolated point $L/2$ to $\{0\}$ lowers gap area from $1/2$ to $1/4$ despite adding zero interval length. Small total measure without anchoring is insufficient.

For a phase ρ and the first output time s_j of column j, call $(S_0, (I_i), E)$ an *admissible cover witness* if $\varnothing \neq S_0 \subseteq \{0, \ldots, L-1\}$, $\kappa_\rho(S_0) < \infty$, all arcs are anchored in S_0, their union has length at most E, and $S_\rho(s_j) \subseteq S_0 \cup \bigcup_i I_i$. Each formal witness supplies a common real subdivision containing the arc endpoints and all raw coordinates, with the latter retaining their values under an order-preserving embedding. Choose any family of such witnesses and set

$$H_{\rho,j} = \sup\left(\{0\} \cup \left\{ \frac{(1-\alpha)_+^2}{2(2K+1)\kappa_\rho(S_0)} - \psi\left(\min\left\{\frac{E}{L}, 1\right\}\right) : \text{chosen admissible witnesses} \right\}\right). \tag{3.6}$$

An empty witness family gives zero. The preceding lemmas show that $0 \leq H_{\rho,j} \leq \mathcal{A}(S_\rho(s_j)) \leq 1/2$, whether or not the supremum is attained. Bounds below that subtract E/L remain valid relaxations of this stronger credit. In particular one may use the exact witness $S_0 = S_\rho(s_j)$, $E = 0$. Let $\overline{H}$ be the arithmetic mean of $H_{\rho,j}$ over the same legal phases and all d columns.

Theorem 3.5 (Theorem C: complete covers with the full paid cost)**.** Lean: `SourceCovers.complete_cover_paid_deficit`, `SourceCovers.paid_deficit_of_gap_credit`, `SourceCovers.phaseCoverCredit_bounds`, `SourceCovers.meanCoverCredit` [`SourceCovers/PaidSourceCover.lean`]

Suppose every actual family outside the selected core and the q exceptions has at most a hops. For every integer $w \geq 1$, define

$$\mathsf{E}_q(w; J, M, R, a, \tau) = \frac{2C_q(T-1)}{dL} + \frac{2(w-1)}{L} + \frac{W-b+1}{2L} + \frac{\beta}{L}\left[J + q + \frac{\tau}{M-R} + \frac{(1+Na)C_q(T-1)}{w}\right]. \tag{3.7}$$

Then the correctness deficit on the fixed full geometry satisfies

$$\frac{\Delta}{L} \geq \overline{H} - \mathsf{E}_q(w; J, M, R, a, \tau). \tag{3.8}$$

All payments in this inequality use the complete bank, with the protected rank phase and the residual grid shift chosen compatibly. The same bound holds with τ replaced by the hop mass of the unprotected selected families that are nonleaves of the original core. The finite statement does not require $T \leq cL$.

Outline of the formal proof. At every step the formal parent relation contains the state predecessor and every actual long work-tape hop; length-one work hops are state edges. The selected relation uses the original core and exceptions, before phase deletion. Pay the distinct outer lowers of those families and all actual lowers deleted by the phase. Across every unpaid selected edge, exact-bank source sets are nested. Every remaining live lower crosses its boundary through a state edge or a hop of an unselected family. Full-span coloring and the outside-family size bound therefore give at most $1 + Na$ such lowers.

Index the exact list bank by its slots. This preserves both its raw value sets and its summed variation, including coincident slots. For each legal phase, Lemma 2.8 supplies one shift whose residual payment is at most $(1 + Na)V_\rho/w$, where V_ρ is that bank's actual raw variation. The whole-record tuple at all distinct paid times is one common transcript for every output. Local determinism and actual output correctness imply functional recovery from this tuple and the complete raw blocks. Thus Theorem 2.9 applies to the same bank and tuple; no global recovery or final payment estimate is assumed.

For each cover witness, telescoping in its supplied real subdivision identifies weighted gap area with the original raw point-gap area. Apply Lemmas 3.3 and 3.4 to bound the witness score by the complete frozen set's gap area. The scores, together with zero, are bounded above by $1/2$, so their supremum satisfies the same bound without an attainment premise. Neither the witness nor its credit depends on the residual grid shift.

Replace the frozen areas by these credits and use $V_\rho \leq C_q(T-1)$. Average each phase's full signed inequality, keeping credit and payment together. The mean initial payment is at most the number of distinct selected outer lowers plus the unprotected nonleaf hop mass divided by the actual number of legal phases. The outer count is at most $J + q$, the hop mass at most τ, and the phase count at least $M - R$. Divide by L to obtain Eqs. (3.7) and (3.8). The formal proof retains the actual variation and payment quantities before these relaxations. □

This theorem separates the two obligations of an impossibility proof: establish a large geometric credit and keep the entire revelation bill below it. Optimizing a witness is an analytic operation on the fixed geometry, not a claim that the machine can compute it cheaply.

Corollary 3.6 (Counting terminal core orbits)**.** Lean: `SourceCovers.phaseCoreCover`, `SourceCovers.meanReciprocalCover`, `SourceCovers.meanReciprocalCover_bounds`, `SourceCovers.reciprocal_cover_paid_deficit`, `SourceCovers.orbit_count_paid_deficit` [`SourceCovers/OrbitCountDeficit.lean`]

Under the same hypotheses, put

$$\kappa_{\rho,j} = \kappa_\rho(S_\rho(s_j)), \qquad \mathcal{R} = \frac{1}{|\mathcal{A}|d} \sum_{\rho\in\mathcal{A}} \sum_{j=0}^{d-1} \frac{1}{\kappa_{\rho,j}},$$

where $\mathcal{A}$ is the set of legal phases and s_j is the first output time of column j. Then

$$\frac{1}{(2M+4)^{|\mathcal{E}|}} \le \mathcal{R} \le 1,$$

$$\frac{\Delta}{L} \ge \frac{(1-\alpha)_+^2}{2(2K+1)}\mathcal{R} - \mathsf{E}_q(w; J, M, R, a, \tau). \tag{3.9}$$

In particular, since $|\mathcal{E}| \le q$,

$$\frac{\Delta}{L} \ge \frac{(1-\alpha)_+^2}{2(2K+1)(2M+4)^q} - \mathsf{E}_q(w; J, M, R, a, \tau). \tag{3.10}$$

Outline of the formal proof. The exact bank at time s_j is a union of at most $(\ell+1)^{|\mathcal{E}|}$ complete core orbits. Each covering root belongs to its own orbit and hence to the bank. Regression puts it at a time at most $s_j < T$, so it is a permitted root in the fractional cover definition. Unit weights give $1 \le \kappa_{\rho,j} \le (2M+4)^{|\mathcal{E}|} < \infty$; the lower bound uses the nonempty complete bank. Taking reciprocals and averaging proves the bounds on $\mathcal{R}$.

Apply Lemma 3.3 to each complete raw bank. Its credit is $(1-\alpha)_+^2/[2(2K+1)\kappa_{\rho,j}]$, the score of the exact zero-arc witness. The formal proof reuses the paid averaging argument of Theorem 3.5, with these gap lower bounds as its credits. Factoring out the common numerator gives Eq. (3.9); the lower bound on $\mathcal{R}$ then gives Eq. (3.10). □

This completes the finite input to the necessary hop-tail theorem below. Retaining the mean reciprocal can improve the finite deficit when actual covers are cheaper than the terminal-orbit count. It preserves variation across columns and phases without assuming economical covers for arbitrary geometries or enlarging the uniform impossibility classes by itself.

3.3 Two-exception cover suppliers

Let g and e be respectively the newer and older exceptional maps, ordered by decreasing owner; tied owners may be ordered either way. If there is just one exception, take it as e and let g be the identity; with no exceptions both maps are the identity. Write

$$R_{e,\rho}(t) = A_\rho(t) \cup \bigcup_{u\in A_\rho(t)} A_\rho(e(u)). \tag{3.11}$$

This is the complete older-exception reachability relation. It is reflexive, transitive and causal. With $Z_\rho(t) = \{g(u) : u \in A_\rho(t),\ g(u) \ne u\}$,

$$O_\rho(t) = R_{e,\rho}(t) \cup \bigcup_{z\in Z_\rho(t)} R_{e,\rho}(z). \tag{3.12}$$

Indeed successful owners decrease strictly, so a path can use g before e but never conversely. Let $F_\rho(t)$ be the maximal elements of $\{t\} \cup Z_\rho(t)$ under the order $u \succeq_e v$ if $v \in R_{e,\rho}(u)$. This is a partial order: a nonidentity step strictly decreases time. The point t is maximal and belongs to $F_\rho(t)$.

Proposition 3.7 (Complete continuation suppliers)**.** Lean: `SourceCovers.oneExceptionBank_exact`, `SourceCovers.twoExceptionBank_exact`, `SourceCovers.continuationSuppliers`, `SourceCovers.continuationSuppliers_exact`, `SourceCovers.continuationSuppliers_exact_bank`, `SourceCovers.continuationSuppliers_root`, `SourceCovers.continuationSuppliers_past`, `SourceCovers.olderLandingSuppliers`, `SourceCovers.olderLandingSuppliers_exact`, `SourceCovers.continuationSuppliers_fractional_bound`, `SourceCovers.olderLandingSuppliers_fractional_bound`, `SourceCovers.continuation_fractional_bounds`, `SourceCovers.clusterVariation`, `SourceCovers.continuationClusters_erosion`, `SourceCovers.continuationClusters_gap`, `SourceCovers.continuationClusters_gap_linear` [`SourceCovers/ContinuationClusters.lean`]

For at most two exceptions the following bounds hold for every legal phase and computation root $t < T$.

(a) $\kappa_\rho(S_\rho(t)) \le \sum_{u \in F_\rho(t)} (1 + |A_\rho(u)|) \le (\ell+1)|F_\rho(t)|$.

(b) Let $C_g(t) = A_\rho(t) \cup \bigcup_{u \in A_\rho(t)} A_\rho(g(u))$. Remove from $\{e(v) : v \in C_g(t),\ e(v) \neq v\}$ the points already in $C_g(t)$, and let $s(t)$ be the number of maximal remaining points under core reachability. Then $\kappa_\rho(S_\rho(t)) \le 1 + |A_\rho(t)| + s(t) \le \ell + 1 + s(t)$.

(c) Cover $F_\rho(t) \setminus \{t\}$ by r closed integer time intervals $I_1, \dots, I_r \subseteq [0, T)$ and choose $a_v \in I_v$. The intervals may overlap, and the representatives need not be suppliers. In particular, the intervals may be the time hulls of a partition into nonempty clusters, with one representative in each cluster. Fix a finite bank of time maps $(b_i)_i$ satisfying $\{b_i(u) : i\} = R_{e,\rho}(u)$ at every time u, as supplied by Theorem 2.7. These geometry-dependent maps are fixed before any residual width or shift is chosen. Put $E(t) = \sum_{v=1}^{r} \sum_i \mathrm{Var}(x \circ b_i; I_v)$. An empty remainder permits $r = E(t) = 0$. Then, with $\psi(e) = e - e^2/2$,

$$\mathcal{A}(S_\rho(t)) \ge \frac{(1-\alpha)_+^2}{2(2K+1)(\ell+1)(r+1)} - \psi\left(\min\left\{\frac{E(t)}{L}, 1\right\}\right).$$

Consequently,

$$\mathcal{A}(S_\rho(t)) \ge \frac{(1-\alpha)_+^2}{2(2K+1)(\ell+1)(r+1)} - \frac{E(t)}{L}. \tag{3.13}$$

Outline of the formal proof. The existing ordered-bank exactness theorem gives both closure decompositions, including identity padding for missing exceptions. A shared finite causal-closure lemma proves that each seed is reached from a maximal seed: among its dominators, choose one of greatest time. Transitivity and regression show that it has no distinct dominator in the seed set. Replacing seeds by these maximal suppliers preserves the whole union of complete closures.

The older closure from u is covered by the core orbit rooted at u and those rooted at $e(v)$ for $v \in A_\rho(u)$. Thus it needs at most $1 + |A_\rho(u)|$ core orbits, proving (a)'s sharper sum. For (b), expand the same ordered bank as $C_g(t)$ followed by all older landings and their complete core continuations. Since $C_g(t)$ is core-closed, landing seeds already in it add nothing. The shared closure lemma reduces the remaining seeds to their core-maximal elements. The newer-only part uses at most $1 + |A_\rho(t)|$ core orbits. In both parts every covering root belongs to the complete source closure, so regression puts it at a legal time below T. Unit weights on these roots are feasible for the fractional minimum. The verified bound $|A_\rho(u)| \le \ell$ gives both original estimates directly.

For (c) use the base raw set $S_0 = x(R_{e,\rho}(t)) \cup \bigcup_v x(R_{e,\rho}(a_v))$. It has a cover by at most $(r+1)(\ell+1)$ complete core orbits, even when representatives repeat or are not in $F_\rho(t)$. For each time interval and older-bank slot, take the minimum and maximum of $x \circ b_i$ on that interval. Both

extrema are actual samples; the existing finite-path variation lemma bounds their difference by $\mathrm{Var}(x \circ b_i; I_v)$. The resulting closed raw interval contains the representative value, an anchor in S_0.

The exact maximal-supplier identity shows that these raw intervals cover every source outside S_0. The proof uses all slots of any chosen exact older bank; repeated slots retain their movement charges. The union of raw intervals has length at most their summed lengths, hence at most $E(t)$. Apply Lemma 3.3 to S_0 and the stronger anchored-erosion bound of Lemma 3.4. Finally $\psi(\min\{E/L, 1\}) \le E/L$ gives Eq. (3.13). No disjointness of the time intervals or unit-speed premise on x is needed for this argument. □

All three statements concern complete continuations after a return. Replacing $x \circ b_i$ by the immediate input coordinate x would remove unaccounted older sources. Actual orbit counts can reduce the fractional cover budget, and the quadratic saving increases the approximation credit for a given movement cost. Allowing interval covers permits suppliers to share an approximation without first assigning them to disjoint groups. These suppliers offer concrete sufficient conditions for Theorem 3.5; no automatic bound on their values for all native geometries is asserted.

4 Uniform exclusions and necessary source complexity

The finite bound has constants independent of the number of selected core families. This permits a numerical limit argument. It does not permit an infinite source bank or a cut parameter growing with dimension.

4.1 Original hop and variation profiles

Fix $c, \beta > 0$, $N \ge 1$ and an integer $q \ge 1$. In each admissible geometry let $\mathcal{K}_0$ be an original Q_2 core and let $\mathcal{E}$ contain at most q other complete families. Let a_0 bound the hop count of every family outside $\mathcal{K}_0 \cup \mathcal{E}$. Sort the original core by nonincreasing hop count, with a fixed rule for ties, and denote it by $G_1, G_2, \ldots$. Both profiles below are zero at indices with no family. The variation profile uses this same ordering, not an ordering by variation.

Definition 4.1 (Near-full sequence and joint profile)**.** Lean: `SourceCovers.NearFullSequence`, `SourceCovers.NearFullSequence.ofGeometry`, `SourceCovers.NearFullSequence.JointProfile`, `SourceCovers.NearFullSequence.exists_jointProfile`, `SourceCovers.NearFullSequence.JointProfile.hop_mass`, `SourceCovers.NearFullSequence.JointProfile.variation_mass`, `SourceCovers.NearFullSequence.JointProfile.hop_antitone`, `SourceCovers.NearFullSequence.JointProfile.hop_le_rank`, `SourceCovers.NearFullSequence.JointProfile.finite_sums_tendsto`, `SourceCovers.NearFullSequence.JointProfile.prefix_add_tail`, `SourceCovers.NearFullSequence.JointProfile.finite_tails_tendsto`, `SourceCovers.NearFullSequence.JointProfile.tails_tendsto_zero` [`SourceCovers/JointProfiles.lean`]

A *near-full sequence* consists of admissible fixed geometries and nonempty correctness sets indexed by ν, with common c, β, N, q, such that

$$d_\nu \longrightarrow \infty, \qquad a_{0,\nu}/L_\nu \longrightarrow 0, \qquad \Delta_\nu/L_\nu \longrightarrow 0. \tag{4.1}$$

Its original profiles are

$$h_i^{(\nu)} = |G_i|/L_\nu, \qquad v_i^{(\nu)} = \mathrm{Var}(x_\nu; [p_{G_i}, U_{G_i}])/L_\nu.$$

A *joint profile* is a coordinatewise subsequential limit $(h_i, v_i)_{i\ge1}$ of both sequences of profiles. Write

$$\tau_R = \sum_{i>R} h_i, \qquad V_K = \sum_{i>K} v_i.$$

Every near-full sequence has a joint profile. For every such profile, both coordinates are nonnegative and summable, with

$$\sum_{i\ge1} h_i \le c, \qquad \sum_{i\ge1} v_i \le 2c, \qquad h_{i+1} \le h_i \le c/i \quad (i \ge 1).$$

Consequently $\tau_R \to 0$ and $V_K \to 0$. Finite coordinate sums converge along the common subsequence; afterwards, finite tails of the limiting profile converge to its infinite tails. None of these statements asserts convergence of the finite geometries' tail masses or stabilization of their family topology.

Outline of the formal proof. Order each geometry's complete core families by nonincreasing hop count, using any fixed tie-breaking rule, and pad both profiles with zero after the last family. The variation profile uses the same list. The formal proof uses Mathlib's list sorting and a sum identity for this enumeration. The Q2 accounting bounds every finite coordinate sum by c for hops and $2c$ for variation, using $T \le cL$.

Place both countable sets of coordinates in one product of compact real intervals. Mathlib's compactness theorem gives one strictly increasing subsequence along which all coordinates converge. For any such limit, nonnegativity and every finite-sum budget pass to the limit. Mathlib's bounded-partial-sum criterion then gives summability and the stated total bounds; it does not exchange an infinite sum with a limit. Hop antitonicity also passes to the limit, and $ih_i \le \sum_{j=1}^{i} h_j \le c$ gives the rank bound.

The verified prefix-plus-tail identities and the convergence of partial sums give the two tail conclusions. Lean indices start at zero: its entry i is the paper's G_{i+1}. Its shift by R therefore corresponds exactly to the paper's tail after the first R families. Strictly increasing subsequences preserve all three near-full conditions. These proofs allow arbitrary positive entry widths, including the logarithmic widths of (T). □

This supplies the common numerical profile used by the next theorem. The next theorem applies the finite deficit to each selected prefix and justifies the successive limits in that inequality.

Theorem 4.2 (Theorem U: necessary hop tails)**.** Lean:
`SourceCovers.NearFullSequence.JointProfile.necessary_hop_tail`,
`SourceCovers.NearFullSequence.JointProfile.tail_floor`,
`SourceCovers.NearFullSequence.JointProfile.not_summable_moment`,
`SourceCovers.NearFullSequence.JointProfile.moment_partial_sums_tendsto_atTop`
[`SourceCovers/NecessaryHopTails.lean`]

For every joint profile of every near-full sequence and all integers $K, R \ge 0$, $M \ge \max(2, R+1)$,

$$\frac{\beta\tau_R}{M-R} \ge \frac{(1-V_K)_+^2}{2(2K+1)(2M+4)^q}. \tag{4.2}$$

Consequently, for every $K, R \ge 0$,

$$\tau_R \ge \frac{(1-V_K)_+^2}{2\beta 4^q(2K+1)(R+2)^{q-1}}. \tag{4.3}$$

If $q \ge 2$, every such profile satisfies $\sum_{i\ge1} i^{q-1}h_i = \infty$.

Outline of the formal proof. The formal library first retains actual cover averages as in Theorem 4.6, then substitutes their terminal-orbit lower bound. We describe the shared finite-prefix calculation here. Fix a joint-profile subsequence and K, R, M. For each fixed integer $J \geq \max(K, R)$, select the first J original core families, distinguish its first K, protect its first R, and rebuild the forest and ranks of that finite selection. These are actual complete families; the formal prefix identities preserve their full-span weights. Distinguishing a family does not require protecting it. Absent families are simply omitted. Sorted hop counts and their total bound imply that every omitted core family has at most $cL_\nu/(J+1)$ hops. The finite theorem therefore applies with

$$a_\nu = \max\{a_{0,\nu}, \lceil cL_\nu/(J+1)\rceil\}, \qquad w_\nu = \lceil \lambda L_\nu \rceil,$$

where $\lambda > 0$ is also fixed before ν tends to infinity. Use the full undistinguished variation and unprotected hop sums as the finite budgets. The latter bounds the nonleaf-only payment after the forest is rebuilt. Their limits are

$$\alpha_\nu = \sum_{K<i\leq J} v_i^{(\nu)} \longrightarrow V_{K,J} := \sum_{K<i\leq J} v_i, \qquad \tau_\nu/L_\nu = \sum_{R<i\leq J} h_i^{(\nu)} \longrightarrow \tau_{R,J} := \sum_{R<i\leq J} h_i.$$

The normalized residual bill is bounded in the limit by $\beta NC_q c^2/[\lambda(J+1)]$; the normalized movement, outer-lower and stripe errors vanish. The width error is at most $2w_\nu/L_\nu \to 2\lambda$. These limits reuse Mathlib's normalized-ceiling and finite-sum results, together with $T_\nu \leq cL_\nu$; no convergence of T_ν/L_ν is required. Using Corollary 3.6 and Eq. (4.1), we obtain

$$\frac{(1-V_{K,J})_+^2}{2(2K+1)(2M+4)^q} \leq \frac{\beta\tau_{R,J}}{M-R} + 2\lambda + \frac{\beta NC_q c^2}{\lambda(J+1)}. \tag{4.4}$$

This is an inequality for every fixed $J \geq \max(K, R)$ and $\lambda > 0$. Now let J tend to infinity in these numerical inequalities, using the verified summability of the limiting profiles. Finally let λ decrease to zero; formally, take $\lambda = \delta/2$ for each $\delta > 0$ and remove the resulting additive δ. The constant C_q depends on M, q, not J, so this proves Eq. (4.2).

Taking $M = 2R+2$, including $R = 0$, gives $2M+4 = 4(R+2)$ and proves Eq. (4.3). A finite K with $V_K \leq 1/2$ always exists. If $\sum_i i^{q-1}h_i$ were finite, then for $R \geq 0$

$$(R+2)^{q-1}\tau_R \leq 3^{q-1}\sum_{i>R} i^{q-1}h_i \longrightarrow 0,$$

contrary to Eq. (4.3) for that K. The shared formal lemma compares scaled tails with a summable majorant and permits any fixed nonnegative shift. Mathlib's nonnegative-series criterion turns the resulting nonsummability into divergence of the partial sums to $+\infty$. Every step used a finite certificate. Omitted families remained in the residual bill until after the dimension limit; no family was discarded by an exchange of limits. □

The independent cutoffs remove the earlier restriction $R \geq K$: the certificate may protect fewer families than it discounts in the variation estimate. This strengthens the necessary profile constraints without by itself enlarging the asymptotic exclusion classes below. The theorem still concerns the specified admissible geometries, not arbitrary tape computations.

Corollary 4.3 (One arbitrary exceptional family)**.** Lean: `SourceCovers.one_exception_uniform_deficit`, `SourceCovers.uniform_deficit_of_no_nearFull`, `SourceCovers.NearFullSequence.one_exception_false`, `SourceCovers.AdmissibleGeometry.deficit_card_iff` [`SourceCovers/UniformDeficit.lean`]

Fix $c, \beta > 0$ and $N \geq 1$. There exist $\varepsilon > 0$ and d_0 such that for every positive entry width b, every admissible geometry with $L = d^2 b$, a Q_2 core, at most one other complete exceptional family, $d \geq d_0$, and $a_0 \leq \varepsilon L$ satisfies

$$\Delta \geq \varepsilon L, \qquad |\mathcal{C}| \leq 2^{(1-\varepsilon)L}. \tag{4.5}$$

The same constants work for all b. No envelope on the sorted core-family sizes is assumed.

Outline of the formal proof. The formal proof first excludes every near-full sequence with $q = 1$. Its joint profile has summable hop mass. The verified critical-moment argument of Theorem 4.2 also applies at $q = 1$, where the weight i^{q-1} is one, and contradicts that summability.

Next apply a shared sequential-exclusion lemma to the two nonnegative costs a_0/L and Δ/L. Nonnegativity of the latter follows from $1 \leq |\mathcal{C}| \leq 2^L$. If no uniform gap existed, for each $n \geq 0$ one could choose an actual admissible geometry with

$$d_n \geq n, \qquad a_{0,n}/L_n \leq \frac{1}{n+1}, \qquad \Delta_n/L_n < \frac{1}{n+1}.$$

Mathlib's reciprocal limit and squeeze theorem give a near-full sequence, contradicting the preceding step. The argument makes no choice of a width function before extracting these geometries, so the constants are uniform over all positive widths. A class-restricted version retains any additional predicate on every selected geometry; it is reusable for the later envelope criteria.

Finally, for nonempty $\mathcal{C}$, Mathlib's base-two logarithm and real-power equivalence gives

$$\varepsilon L \leq \Delta \quad \Longleftrightarrow \quad |\mathcal{C}| \leq 2^{(1-\varepsilon)L}.$$

Thus both conclusions use the same constants. The Lean statement also covers nonpositive ambient capacity bounds: the necessary-tail inequality then contradicts the positive geometric credit directly. □

This verifies a uniform exclusion for the stated return geometries, including bit matrices and logarithmic-size entries. It does not bound general transposers without the full-span partition and core hypotheses. The proof gives existence of uniform constants, not numerical values.

Corollary 4.4 (Common size envelopes)**.** `Lean: SourceCovers.EnvelopeTailSmall, SourceCovers.finite_prefix_envelope_uniform_deficit, SourceCovers.common_envelope_uniform_deficit, SourceCovers.power_envelope_tailSmall, SourceCovers.power_envelope_uniform_deficit` [`SourceCovers/CommonEnvelopes.lean`]

Fix c, β, N, q as above and a nonnegative summable sequence (g_i) such that

$$\liminf_{R \to \infty} (R+2)^{q-1} \sum_{i > R} g_i = 0. \tag{4.6}$$

Write $s_i = |G_i|/L$ in the original hop-sorted order, with $s_i = 0$ after the last family. There exist $\varepsilon > 0$ and an integer $m \geq 0$ such that every admissible geometry satisfying

$$d \geq m, \qquad a_0 \leq \varepsilon L, \qquad s_i \leq g_i + \varepsilon \quad (1 \leq i \leq m)$$

satisfies Eq. (4.5). The constants are common to all positive entry widths. No envelope condition is imposed beyond rank m. In particular, the conclusion holds under the original exact common bound $|G_i| \leq L g_i$ on all core families. Every common power envelope $g_i = H/i^p$, with fixed $H \geq 0$ and real $p > q$, satisfies Eq. (4.6) and is covered.

Outline of the formal proof. First suppose a joint profile obeys $h_i \le g_i$ for every i. Both sequences are summable, so their tails satisfy $\tau_R \le \sum_{i>R} g_i$. Choose K with $V_K < 1$ in Eq. (4.3). It gives a constant $A > 0$ such that

$$A \le (R+2)^{q-1}\tau_R \le (R+2)^{q-1} \sum_{i>R} g_i \quad \text{for every } R \ge 0,$$

contradicting Eq. (4.6). The lower limit is taken in $[0, +\infty]$: being zero means that for every $\eta > 0$ and every R_0, some $R \ge R_0$ has scaled tail below η. The formal proof uses this equivalent condition, which needs no upper bound on the scaled tails.

To obtain the stronger finite-prefix statement, consider an actual admissible geometry together with a cutoff J, a slack $\eta \ge 0$, and one sorted order satisfying $s_i \le g_i + \eta$ for $1 \le i \le J$. Apply the shared uniform-gap lemma to effective dimension $\min(d, J)$ and the two costs

$$\max\{a_0/L, \eta\}, \qquad \Delta/L.$$

If effective dimension tended to infinity and both costs tended to zero, the geometries would form a near-full sequence with $J \to \infty$ and $\eta \to 0$. Retain each supplied order for both numerical profiles. Along their common subsequence every fixed rank eventually lies below the cutoff, hence $h_i \le g_i$ by passage to the limit. The preceding contradiction excludes such a sequence. The uniform-gap lemma therefore provides ε, m; taking $J = m$ and $\eta = \varepsilon$ proves the claimed deficit, and the verified logarithmic equivalence gives its cardinality form. The exact-envelope conclusion follows by adding the positive slack. Only one order is supplied per geometry; no additional requirement on tie breaking is introduced.

For $g_i = H/i^p$, Mathlib's p-series theorem gives

$$\sum_{i \ge 1} i^{q-1} g_i = H \sum_{i \ge 1} i^{q-1-p} < \infty \qquad (p > q).$$

The existing moment-to-tail lemma then gives $(R+2)^{q-1} \sum_{i>R} g_i \to 0$, which implies Eq. (4.6). This reuses the weighted-tail argument of Theorem 4.2 without a separate integral estimate. Lean's rank i corresponds to the paper's rank $i+1$ throughout. □

Thus ruling out a common limiting envelope also rules out a finite set of approximate leading-rank bounds. At sufficiently large dimension, a construction in this admissible class must violate at least one such bound if its outside families and correctness deficit are both small. The hypotheses on the full geometry remain necessary; the constants and cutoff are not given numerically.

Corollary 4.5 (A fixed finite skeleton)**.** Lean: `SourceCovers.AdmissibleGeometry.withSkeleton`, `SourceCovers.NearFullSequence.empty_core_false`, `SourceCovers.empty_core_uniform_deficit`, `SourceCovers.finite_skeleton_uniform_deficit`, `SourceCovers.large_family_count_of_small_deficit` [`SourceCovers/FiniteSkeleton.lean`]

Fix $c, \beta > 0$, $N \ge 1$ and an integer $q \ge 0$. There are $\varepsilon > 0$ and d_0 such that Eq. (4.5) holds for every admissible geometry with $d \ge d_0$, any set $\mathcal{E}$ of at most q complete families, and any integer $a \ge 0$ satisfying

$$|G| \le a \quad (G \notin \mathcal{E}), \qquad a \le \varepsilon L.$$

The chosen skeleton $\mathcal{E}$ need not agree with the geometry's previous core or exception labels. The constants are common to all positive entry widths and all previous classifications. Equivalently, with these constants, $d \ge d_0$ and $\Delta < \varepsilon L$ force

$$\#\{G : \ |G| > \varepsilon L\} > q,$$

where the count ranges over all complete families of the computation.

Outline of the formal proof. Reclassify the complete families by taking the core empty, the exceptions to be $\mathcal{E}$, and the outside bound to be a. The full work walks, raw path, output times, records, parent relation, correctness set and N-class span partition are unchanged. The supplied size hypothesis justifies the new outside bound; no family or hop is deleted.

A hypothetical near-full sequence with empty cores has an explicit zero joint profile, using the identity subsequence. The verified zero-padding lemmas give both coordinates. The verified form of Eq. (4.2) also permits $q = 0$. At $K = R = 0$ and $M = 2$, it becomes $1/(2 \cdot 8^q) \le 0$, a contradiction for every $q \ge 0$. The existing uniform-gap theorem gives the empty-core deficit, and reclassification transfers it to the proposed skeleton. The constants are chosen before the original classification or entry width. The Lean proof also permits nonpositive ambient capacity bounds.

For the counting form, take $\mathcal{E}$ to be the families with $|G| > \varepsilon L$ and $a = \lfloor \varepsilon L \rfloor$. If $|\mathcal{E}| \le q$, Mathlib's natural-floor bounds give the preceding hypotheses, contradicting $\Delta < \varepsilon L$. Conversely, the count precludes any proposed skeleton of size at most q when all remaining families have at most $a \le \varepsilon L$ hops. □

Thus a construction with sufficiently small correctness deficit needs more than any prescribed fixed number q of families of size proportional to L, with the proportionality constant allowed to depend on q. This conclusion retains the full-span partition hypothesis.

These are restrictions on a common full geometry. In the one-exception case they settle the paid-deficit comparison without a coding conjecture. For $q = 2$, the necessary tail floor is of order $1/R$, compatible with an inverse-square size profile; a further geometric argument is needed. The proof supplies uniform constants by contradiction, not effective numerical values of ε and d_0.

4.2 Necessary fractional-cover complexity

Along a fixed joint-profile subsequence, select the first J original core families, protect its first R, and fix $M \ge \max(2, R+1)$. Rebuild the forest and phases for each selected prefix. For every legal phase and column put $\kappa_{\nu,J,R,M,\rho,j} = \kappa_\rho(S_\rho(s_j))$. These are the fractional covers of the complete selected source banks; they are finite and at least one. Define

$$Q^*_{R,M} = \limsup_{J\to\infty} \limsup_{\nu\to\infty} \mathbb{E}_{\rho,j} \frac{1}{\kappa_{\nu,J,R,M,\rho,j}}, \qquad Q^*_R = Q^*_{R,2R+2}. \tag{4.7}$$

The expectation is the arithmetic mean over legal phases and columns. Both limits hold R, M fixed. We suppress $M = 2R+2$ in the subscript of κ when using Q^*_R.

Theorem 4.6 (Necessary reciprocal-cover bound)**.** Lean:
`SourceCovers.NearFullSequence.JointProfile.reciprocalCoverLimit`,
`SourceCovers.NearFullSequence.JointProfile.reciprocalCoverLimit_bounds`,
`SourceCovers.NearFullSequence.JointProfile.weighted_reciprocal_cover`,
`SourceCovers.NearFullSequence.JointProfile.necessary_reciprocal_cover`,
`SourceCovers.NearFullSequence.JointProfile.necessary_reciprocal_cover_default`,
`SourceCovers.NearFullSequence.JointProfile.coverProbabilityLimit`,
`SourceCovers.NearFullSequence.JointProfile.coverProbabilityLimit_le_reciprocal`,
`SourceCovers.NearFullSequence.JointProfile.hop_le_of_pointwise`,
`SourceCovers.NearFullSequence.JointProfile.inverse_square_cover_probability`
[`SourceCovers/NecessaryCover.lean`]

For every joint profile of every near-full sequence, all integers $K, R \ge 0$, and every $M \ge \max(2, R+1)$,

$$\frac{(1 - V_K)^2_+}{2(2K+1)} Q^*_{R,M} \le \frac{\beta \tau_R}{M - R}. \tag{4.8}$$

If $V_K < 1$, this gives

$$Q^*_{R,M} \le \frac{2\beta(2K+1)\tau_R}{(1-V_K)^2(M-R)}.$$

In particular,

$$Q^*_R \le \frac{2\beta(2K+1)\tau_R}{(1-V_K)^2(R+2)}. \tag{4.9}$$

The cutoffs K, R are independent, including $R = 0$, and the statements hold for any fixed number q of exceptions. If the limiting profile satisfies $h_i \le H/i^2$ for some $H \ge 0$ and $V_K \le 1/2$, then for every $R \ge 1$ and every fixed real $s \ge 0$,

$$\limsup_J \limsup_\nu \Pr_{\rho,j}\{\kappa_{\nu,J,R,\rho,j} \le s\} \le \frac{8\beta(2K+1)Hs}{R(R+2)}. \tag{4.10}$$

A common bound $|G_i| \le HL_\nu/i^2$ in the supplied original hop order is sufficient for this limiting-envelope hypothesis.

Outline of the formal proof. Retain the actual mean reciprocal cover in the finite paid-deficit inequality. The prefix and rounding estimates described in the proof of Theorem 4.2 are proved once in this stronger form. For fixed $J \ge \max(K, R)$, M and $\lambda > 0$, they give

$$\frac{(1-V_{K,J})_+^2}{2(2K+1)} \limsup_\nu \mathbb{E}_{\rho,j} \frac{1}{\kappa_{\nu,J,R,M,\rho,j}} \le \frac{\beta\tau_{R,J}}{M-R} + 2\lambda + \frac{\beta N C_q c^2}{\lambda(J+1)}.$$

The finite averages and their successive upper limits lie in $[(2M+4)^{-q}, 1]$. Thus real upper limits are bounded throughout. A shared lemma uses Mathlib's subsequence attaining the upper limit to pass a converging coefficient through this inequality. It also handles a zero limiting coefficient; no division is made at this stage.

Apply the same lemma as J tends to infinity. Summability gives $V_{K,J} \to V_K$ and $\tau_{R,J} \to \tau_R$, while the residual error tends to zero. Finally take $\lambda = \delta/2$ for every $\delta > 0$ and remove the additive δ. This proves Eq. (4.8). When $V_K < 1$, division by its positive coefficient gives the general bound, and $M = 2R + 2$ gives Eq. (4.9). The dimension limit, prefix limit and removal of the width error occur in that order.

For the probability bound, $\kappa \ge 1$ gives the finite pointwise inequality $\mathbf{1}_{\{\kappa \le s\}} \le s/\kappa$ for every $s \ge 0$. Averaging and taking the same two upper limits preserves it. Mathlib's finite inverse-square sum estimate, followed by the summable limit, gives $\tau_R \le H/R$ when $R > 0$. The bound $(1-V_K)^2 \ge 1/4$ then gives Eq. (4.10). Only the limiting envelope is used. The stated common finite envelope passes coordinatewise to the same joint profile. □

This strengthens the necessary condition by retaining the actual cover complexity: replacing $Q^*_{R,M}$ by its terminal-orbit lower bound $(2M+4)^{-q}$ recovers Eq. (4.2) directly. It does not prove that arbitrary admissible geometries have small covers. In the inverse-square case it limits the fraction of phase-column pairs with small cover, not only their mean cover complexity.

Corollary 4.7 (Uniform cover criteria)**.** Lean:
`SourceCovers.AdmissibleGeometry.HasMeanCoverBound`, `SourceCovers.CoverTailSmall`, `SourceCovers.uniform_deficit_of_no_ordered_nearFull`, `SourceCovers.linear_mean_cover_uniform_deficit`, `SourceCovers.envelope_mean_cover_uniform_deficit`, `SourceCovers.power_cover_tailSmall`, `SourceCovers.power_mean_cover_uniform_deficit` [`SourceCovers/UniformCover.lean`]

Fix $c, \beta > 0$, $N \ge 1$ and an integer $q \ge 0$. Each of the following defines a class with constants $\varepsilon > 0, d_0$ for which Eq. (4.5) holds when $d \ge d_0$ and $a_0 \le \varepsilon L$, uniformly over all positive entry widths.

(a) For some common constant $B \geq 0$, selecting the first J original core families and protecting the first R, for every $J \geq R \geq 1$ and $M = 2R + 2$, gives $\mathbb{E}_{\rho,j}\kappa_{J,R,\rho,j} \leq B(R+2)$. No size envelope is required.

(b) For fixed $H \geq 0$ and real $p > 1$, the original sizes satisfy $|G_i| \leq HL/i^p$, and for a common nonnegative function $f(R) = o(R^p)$ every such finite prefix selection satisfies $\mathbb{E}_{\rho,j}\kappa_{J,R,\rho,j} \leq f(R)$.

More generally, in (b) one may replace the power envelope by any common nonnegative summable envelope $|G_i| \leq Lg_i$, and the growth condition by

$$\liminf_{R\to\infty} \frac{f(R)}{R+2} \sum_{i>R} g_i = 0, \tag{4.11}$$

where the lower limit is taken in $[0, +\infty]$. The size and mean-cover bounds must hold in the same supplied original hop order. The constants are independent of the choice of that order. Here $\kappa_{J,R,\rho,j}$ is the fractional cover of the selected geometry's complete source set at the first output time of column j. The bounds are uniform over the class and dimension; ε, d_0 may depend on all the fixed class data. In (b), the inverse-square case requires mean cover $o(R^2)$.

Outline of the formal proof. The formal finite averaging lemma applies Mathlib's Cauchy inequality to the actual legal phase-column pairs. Their cover weights are finite and at least one, so

$$\mathbb{E}_{\rho,j}\frac{1}{\kappa} \geq \frac{1}{\mathbb{E}_{\rho,j}\kappa} \geq \frac{1}{f(R)}$$

whenever the stated mean bound holds. Positivity of $f(R)$ for $R \geq 1$ follows from that bound itself. Holding R fixed, pass first through the dimension upper limit, then through the prefix upper limit. Only the prefixes $J \geq R$ matter for the latter. The explicit bounds from Theorem 4.6 justify both real upper limits. Its weighted inequality therefore gives, for every $K \geq 0$ and $R \geq 1$,

$$A_K := \frac{(1 - V_K)_+^2}{2(2K+1)} \leq \beta \frac{\tau_R f(R)}{R+2}.$$

Choose K with $V_K < 1$, so $A_K > 0$. In (a), the right side is $\beta B \tau_R \to 0$, a contradiction. Under a common summable envelope, coordinatewise convergence gives $h_i \leq g_i$ and summability gives $\tau_R \leq \sum_{i>R} g_i$. Condition Eq. (4.11) provides arbitrarily late values below every positive threshold, again contradicting $A_K > 0$. No boundedness of the scaled envelope tails is assumed.

For the power case, Mathlib's integral comparison and improper-integral formula give, for every $R \geq 1$,

$$\sum_{i>R} \frac{H}{i^p} \leq \frac{H}{p-1} R^{1-p}.$$

Consequently the expression in Eq. (4.11) is at most $Hf(R)/[(p-1)R^p] \to 0$. The shared formal result allows every real $p > 1$ and compares genuinely summable series before taking their tails.

Finally apply the existing uniform-gap theorem to the class of actual geometries together with an order witnessing the required bounds. The adapter retains that same order when forming a near-full sequence and extracting its joint profile. The contradictions above exclude such a sequence, giving uniform ε, d_0. The existing deficit/cardinality equivalence yields both conclusions in Eq. (4.5) with the same constants. No width function is fixed during this extraction. □

The general criterion covers non-power-law and irregular envelopes; only arbitrarily late small values of the weighted tail are required. A proposed construction in this admissible class must therefore violate the corresponding mean-cover budget if its outside families and correctness deficit are sufficiently small. The result does not provide those budgets for arbitrary tape computations or numerical gap constants.

The saturation bound also rules out a misleading average picture: in a near-full inverse-square sequence, rare very complicated columns cannot compensate for a positive fraction with cover complexity $o(R^2)$, in the stated ordered limits. This is a necessary condition, not a construction of an economical cover.

4.3 The movement of complete older observables

For $q = 2$, use the fixed exact older bank constructed in Theorem 2.7, including every slot and its multiplicity. At the first output time s_j of each column, take the time hull of $F_\rho(s_j) \setminus \{s_j\}$ in Proposition 3.7(c). For an empty set take the degenerate interval $[s_j, s_j]$; both an empty set and a singleton have movement $E = 0$. For a fixed modulus $M \geq 2$ and protected cutoff $R < M$, define

$$e_{\nu,J,R,M} = \mathbb{E}_{\rho,j} \frac{E(s_j)}{L_\nu}, \qquad \pi_{\nu,J,R,M} = \mathbb{E}_{\rho,j} \psi\left(\min\left\{\frac{E(s_j)}{L_\nu}, 1\right\}\right), \qquad \psi(z) = z - z^2/2,$$

where the averages use the admissible phases and all columns for the first J original core families. Put $e^*_{R,M} = \liminf_J \liminf_\nu e_{\nu,J,R,M}$ and $\pi^*_{R,M} = \liminf_J \liminf_\nu \pi_{\nu,J,R,M}$, on the same joint-profile subsequence. These are finite real lower limits: $0 \leq e_{\nu,J,R,M} \leq cC_1$ and $0 \leq \pi_{\nu,J,R,M} \leq 1/2$, where C_1 is the one-exception bank coefficient of Eq. (2.9) at the fixed modulus M.

Corollary 4.8 (Necessary complete-observable movement)**.** Lean: `SourceCovers.phaseMovement`, `SourceCovers.NearFullSequence.JointProfile.movementLimit`, `SourceCovers.NearFullSequence.JointProfile.completeMovement_bounds`, `SourceCovers.NearFullSequence.JointProfile.necessary_movement_penalty`, `SourceCovers.NearFullSequence.JointProfile.movementPenalty_le_complete`, `SourceCovers.NearFullSequence.JointProfile.necessary_complete_movement_default`, `SourceCovers.NearFullSequence.JointProfile.scaled_complete_movement_liminf` [`SourceCovers/NecessaryMovement.lean`]

Along every subsequence realizing a joint profile of a two-exception near-full sequence, for all integers $K, R \geq 0$ and $M \geq 2$ with $R < M$,

$$e^*_{R,M} \geq \pi^*_{R,M} \geq \frac{(1 - V_K)^2_+}{4(2K+1)(2M+4)} - \frac{\beta\tau_R}{M - R}.$$

In particular, set $e^*_R = e^*_{R,2R+2}$. For all $K, R \geq 0$,

$$e^*_R \geq \frac{(1 - V_K)^2_+}{16(2K+1)(R+2)} - \frac{\beta\tau_R}{R+2}. \tag{4.12}$$

For fixed K with $V_K < 1$ this implies $\liminf_{R\to\infty}(R+2)e^*_R \geq (1 - V_K)^2/[16(2K+1)]$, where this last lower limit is in $[0, \infty]$.

Outline of the formal proof. Sort the actual exceptions by decreasing owner, using the identity in the newer slot when only one exception is present. The verified ordered-bank identity identifies this representation with the complete bank used for the paid theorem. Apply Proposition 3.7(c) with one interval and its left endpoint as representative. The minimum and maximum supplier

times determine the interval; a degenerate interval handles the empty set. The resulting credit is $(1-\alpha)_+^2/[4(2K+1)(2M+4)]$ minus the quadratic penalty. The whole older bank has movement at most $C_1(T-1)$, so $T \le cL$ gives the stated uniform bounds on the means at fixed M.

Insert this credit into the same paid-deficit theorem. Finite expectation identities leave exactly the mean penalty on the other side, with the original residual error unchanged. A shared error-limit lemma chooses the rounded width $w = \lceil \lambda L \rceil$ and omitted-family budget and proves the error limit independently of the chosen geometric credit. The reciprocal-cover proof and the present proof both reuse it. Pass first to the dimension lower limit, then to the prefix lower limit, using only the common joint profile and explicit boundedness. Eventually $J \ge \max(K, R)$, so no comparison of K and R is needed. Finally let λ decrease to zero.

Since $\psi(\min\{z, 1\}) \le z$ for $z \ge 0$, comparison before each lower limit gives the linear-movement conclusion directly. Set $M = 2R + 2$ and then multiply by $R + 2$. The vanishing tail $\tau_R \to 0$ gives the last assertion; the Lean proof uses an extended-nonnegative lower limit to include possible divergence to infinity. □

Here the moving observables include every older continuation, not just the immediate input coordinate at an exceptional landing. The quadratic bound caps each phase–column pair's penalty at $1/2$; arbitrarily large movement on that pair makes no further contribution to this mean. For fixed K with $V_K < 1$, the linear bound excludes $e_R^* = o(1/R)$. A successful approximation must obtain enough gap credit while paying this complete movement. The result does not establish the smaller movement estimates needed to contradict arbitrary native geometries.

5 Paying shared raw support once

The preceding covers estimate source gap area without adding revelations. Here we use a different mechanism: reveal a raw source set shared by many outputs, pay its union once, and apply the one-exception area bound to the sources that remain. All assertions in this section use $q = 2$.

5.1 A finite global-sharing bound

Retain the notation A_ρ, $R_{e,\rho}$ and $Z_\rho(t)$ of Eqs. (3.11) and (3.12). Fix $J \ge R \ge \max(K, 1)$, $M = 2R + 2$, and set

$$\ell = 4R + 7, \quad C_R = C_0(1 + 3C_0)^2, \quad B_R = \ell(\ell + 1), \quad A_R = \beta N C_R c, \qquad C_0 = 16 + 24M(M-1). \tag{5.1}$$

Let $\mathcal{T}_{\mathrm{out}}$ be the set of all L actual output-bit times, and define

$$Z_\rho^{\mathrm{all}} = \bigcup_{t \in \mathcal{T}_{\mathrm{out}}} Z_\rho(t), \qquad G_\rho = \left| \max_{\succeq_e} Z_\rho^{\mathrm{all}} \right|, \qquad \overline{G}_{J,R} = \mathbb{E}_\rho G_\rho.$$

An empty seed set has $G_\rho = 0$. The order $\succeq_e$ is complete older reachability, not merely temporal order. These maximal seeds are geometric witnesses, not independent records of bounded bit capacity. Let

$$S_\rho^{\mathrm{new}} = x\left(\bigcup_{z \in Z_\rho^{\mathrm{all}}} R_{e,\rho}(z) \right). \tag{5.2}$$

Transitivity and maximality cover this entire set by the older closures of the G_ρ maximal seeds. Each has at most B_R endpoints, so $|S_\rho^{\mathrm{new}}| \le B_R G_\rho$.

Proposition 5.1 (The full shared-support payment)**.** Lean:
`SourceCovers.shared_support_count_deficit`, `SourceCovers.shared_support_paid_deficit`, `SourceCovers.phase_shared_deficit`, `SourceCovers.sharedRawSupport`, `SourceCovers.sharedSeedCount`, `SourceCovers.exactSourceBank_shared_cover`, `SourceCovers.sharedRawSupport_card_le_sum`, `SourceCovers.sharedRawSupport_blocks_card_le`, `SourceCovers.exists_paidBank_shared_deficit` [SourceCovers/GlobalSharing.lean]

With α, τ, a as in Section 3, for every integer $w \geq 1$,

$$\frac{\Delta}{L} \geq \frac{(1-\alpha)_+^2}{2(2K+1)(\ell+1)} - \frac{wB_R\overline{G}_{J,R}}{L} - \mathsf{E}_2(w; J, 2R+2, R, a, \tau). \tag{5.3}$$

The additional raw revelation has at most wB_RG_ρ bits for every grid shift, so the residual-cut shift may be chosen using the original complete two-exception bank.

More sharply, let $\overline{S} = \mathbb{E}_\rho|S_\rho^{\mathrm{new}}|$ and $C_1 = C_0(1+3C_0)$. Define $\mathsf{E}_{1|2}$ by replacing only the first movement term $2C_R(T-1)/(dL)$ in E_2 by $2C_1(T-1)/(dL)$; its residual-cut term still uses C_R. Then

$$\frac{\Delta}{L} \geq \frac{(1-\alpha)_+^2}{2(2K+1)(\ell+1)} - \frac{w\overline{S}}{L} - \mathsf{E}_{1|2}.$$

Neither finite bound requires a linear-runtime hypothesis. The formal statements also allow any $M \geq 2$ with $R < M$, taking $\ell = 2M+3$; the distinguished-family budget K and protection budget R may be independent.

Outline of the formal proof. Fix a phase and select its residual shift θ for the complete bank. In fixed raw-coordinate order reveal every bit in

$$U_{\rho,\theta} = \bigcup_{y \in S_\rho^{\mathrm{new}}} B_\theta(y).$$

For every θ, the finite-union and same-block cardinality bounds give

$$|U_{\rho,\theta}| \leq \min\{L, w|S_\rho^{\mathrm{new}}|\} \leq wB_RG_\rho.$$

Complete older closure is unchanged when its seeds are replaced by maximal seeds, so the raw-support count also retains the sum of their actual closure sizes before using B_RG_ρ. The source-replacement lemma applies whenever the full bank's sources are contained in the remaining bank's sources and a common raw set. Equal original transcripts, equal revealed raw bits, and equal remaining source blocks then imply equal full-bank source blocks. Applying the original recovery theorem therefore gives every output from the older bank and one common joint transcript on the same correctness set. Its frozen source gap has the one-exception lower bound $(1-\alpha)_+^2/[2(2K+1)(\ell+1)]$ by Lemma 3.3. Use the older bank for these sources; its summed variation is at most $C_1(T-1)$, while the full-bank variation in the residual-cut payment is at most $C_R(T-1)$. Theorem 2.9 applies with total information charge $\beta|\Gamma| + |U_{\rho,\theta}|$, without independence assumptions. The intermediate formal bound keeps the actual union cardinality and both actual variations. Average the complete signed inequality over the same phases and use the existing average rank-payment bound. This gives the sharper statement; $\overline{S} \leq B_R\overline{G}_{J,R}$ and $C_1 \leq C_R$ give Eq. (5.3) directly. Revealing a seed vertex itself would not reveal its entire source closure; the raw bits above are what have actually been paid for. □

This supplies a verified finite certificate when many outputs share complete raw support. It does not assert that such support is small in an arbitrary tape computation; the following criteria isolate the quantitative bounds that would suffice.

5.2 Uniform sharing criteria

For the uniform statements below, select the first J original core families in the hop ordering of Section 4, protect the first R, and rebuild the selected core's forest and ranks before choosing a phase. All quantities indexed by J, R, including $\overline{G}_{J,R}$ and the raw covering numbers below, refer to these prefix selections. Absent families are omitted.

Let $(\eta_J)_{J\geq 1}$ be a fixed positive sequence tending to zero that bounds the normalized size of the first omitted original core family: $|G_{J+1}|/L \leq \eta_J$, uniformly over the class. The universal choice $\eta_J = c/(J+1)$ is available from sorted hop counts. Whenever another bound is given, it may be replaced by its minimum with $c/(J+1)$.

Theorem 5.2 (Theorem H: uniform exclusion by global sharing)**.** `Lean`: `SourceCovers.sharing_uniform_deficit`, `SourceCovers.liminf_sharing_uniform_deficit`, `SourceCovers.sublinear_sharing_uniform_deficit`, `SourceCovers.power_sharing_uniform_deficit`, `SourceCovers.AdmissibleGeometry.HasSharingBounds`, `SourceCovers.NearFullSequence.JointProfile.prefix_sharing_limit`, `SourceCovers.NearFullSequence.JointProfile.sharing_tail_lower` [`SourceCovers/UniformSharing.lean`]

Fix c, β, N, a common sequence η_J as above, and an unbounded set $\mathcal{R} \subseteq \{1, 2, \ldots\}$. For each $R \in \mathcal{R}$ fix a finite nonnegative function $g_R(J)$ on $J \geq R$ such that

$$\liminf_{J\to\infty} \eta_J g_R(J) = 0. \tag{5.4}$$

Consider admissible two-exception geometries for which $\overline{G}_{J,R} \leq g_R(J)$ for every $R \in \mathcal{R}$ and $J \geq R$. These bounds and index sets are fixed independently of dimension. There exist $\varepsilon > 0, d_0$ such that every such geometry with $d \geq d_0$ and $a_0 \leq \varepsilon L$ satisfies $\Delta \geq \varepsilon L$, equivalently $|\mathcal{C}| \leq 2^{(1-\varepsilon)L}$. The same conclusion holds for nonnegative η_J tending to zero.

In particular it suffices, without a size envelope, that $\liminf_J g_R(J)/(J+1) = 0$ for each $R \in \mathcal{R}$. Under $|G_i| \leq HL/i^p$ with fixed $H \geq 0$ and $p > 0$, it suffices that $\liminf_J g_R(J)/(J+1)^p = 0$; this includes subquadratic sharing for the inverse-square envelope.

Outline of the formal proof. Negating the conclusion gives a near-full sequence with a joint profile. Choose a finite K with $V_K < 1$ and fix $R \in \mathcal{R}$. For each finite $J \geq R$ use $a = \max(a_0, \lceil \eta_J L \rceil)$ and $w = \lceil \lambda L \rceil$, with $\lambda > 0$ fixed before dimension. Take $J \geq K$ as well. The finite sharing bound permits independent K, R and any positive integer width. Its dimension limit gives

$$\frac{(1 - V_{K,J})^2}{2(2K+1)(\ell+1)} - \frac{\beta \tau_{R,J}}{R+2} \leq \lambda[2 + B_R g_R(J)] + \frac{A_R \eta_J}{\lambda}. \tag{5.5}$$

The finite sums $V_{K,J}, \tau_{R,J}$ are as in Eq. (4.4), and $V_{K,J} \leq V_K < 1$. Let $\mu_{J,R}$ be the left side and put $D_J = 3 + B_R g_R(J) > 0$. Increase the coefficient 2 to 3. Whenever $\mu_{J,R} > 0$, test at

$$\lambda = \frac{\mu_{J,R}}{2D_J} \quad \text{to obtain} \quad \mu_{J,R}^2 \leq 4 A_R \eta_J D_J.$$

This uses no positive lower bound on A_R or η_J and no square-root minimizer. The right side tends to zero along arbitrarily late cutoffs provided by Eq. (5.4), since $\eta_J \to 0$. The finite sums converge along those cutoffs, so their limiting credit cannot be positive. As $\ell + 1 = 4(R+2)$, this gives

$$\beta \tau_R \geq \frac{(1 - V_K)^2}{8(2K+1)}.$$

This holds for every $R \in \mathcal{R}$, contradicting $\tau_R \to 0$. The two special cases use respectively $\eta_J = c/(J+1)$ and $\eta_J = H/(J+1)^p$. Any $p > 0$ makes the latter tend to zero; summability of the actual profile was already proved and need not follow from the envelope. The formal lower limits take values in $[0, \infty]$; zero is equivalent to arbitrarily late values below every positive threshold. □

The bound g_R may grow arbitrarily fast with the fixed protected scale R. Its uniformity in dimension is essential: for a single finite geometry, increasing J beyond all its families proves no uniform statement. A common unbounded set of cutoffs J for each R also suffices if the stated bounds and smallness hold on that set; this is included in the formal statement. The omitted-size bound is then needed only at those same cutoffs. For $0 < p \leq 1$, the power-law case is already covered by the no-envelope criterion; the broader exponent range removes an unnecessary restriction rather than adding a new obstruction in that range. The theorem requires these uniform sharing bounds as hypotheses; it does not establish them for arbitrary tape computations.

Corollary 5.3 (Necessary global continuation density)**.** Not formalized; a complete proof is given below. *Fix a near-full sequence with common omitted-size bound η_J, and restrict it to a subsequence realizing a joint profile. All dimension limits in this statement are along that subsequence. For fixed K with $V_K < 1$ and $R \geq \max(K, 1)$, define*

$$\mu_R = \frac{(1 - V_K)^2}{8(2K+1)(R+2)} - \frac{\beta \tau_R}{R+2}, \qquad g_{J,R}^{\mathrm{low}} = \liminf_{\nu \to \infty} \overline{G}_{J,R}^{(\nu)}.$$

If $\mu_R > 0$, then

$$\liminf_{J \to \infty} \eta_J g_{J,R}^{\mathrm{low}} \geq \frac{\mu_R^2}{4 A_R B_R}. \tag{5.6}$$

The condition $\mu_R > 0$ holds for all sufficiently large fixed R. For $\eta_J = c/(J+1)$ this forces positive linear growth of $g_{J,R}^{\mathrm{low}}$ in J; an inverse-square omitted-size bound forces positive quadratic growth.

Proof (not formalized). For each fixed J with finite $g_{J,R}^{\mathrm{low}}$, pass to a dimension subsequence realizing that liminf. The joint profile is unchanged. Equation (5.5) holds with $g_R(J)$ replaced by this liminf. Its left side $\mu_{J,R}$ tends to μ_R as J grows and is positive for all sufficiently large J. Testing at $\lambda = \mu_{J,R}/[2(2 + B_R g_{J,R}^{\mathrm{low}})]$ and rearranging gives

$$\eta_J g_{J,R}^{\mathrm{low}} \geq \frac{\mu_{J,R}^2}{4 A_R B_R} - \frac{2 \eta_J}{B_R}.$$

Infinite $g_{J,R}^{\mathrm{low}}$ satisfies the inequality automatically. Take the liminf in J. Finally $\tau_R \to 0$ shows that μ_R is positive for all sufficiently large R. □

This is a count of globally incomparable complete continuations. It is not a count of independent raw bits or a native gate lower bound.

5.3 Raw metric covers instead of temporal counts

For $0 < \delta < 1$, let $\mathcal{N}_{\rho,J,R}(\delta)$ be the minimum number of closed real intervals of length at most δL covering S_ρ^{new}. An empty set has covering number zero. These are intervals in the linear raw-coordinate range, not circular arcs.

Proposition 5.4 (Metric shared-support criterion)**.** Not formalized; a complete proof is given below.
In the setting of Theorem 5.2, replace its bounds on $\overline{G}_{J,R}$ by the following. For each R in a fixed unbounded set, there are dimension-independent numbers $\delta_{J,R} \in (0,1)$ and $n_{J,R} \geq 0$ with

$$\mathbb{E}_\rho \mathcal{N}_{\rho,J,R}(\delta_{J,R}) \leq n_{J,R}.$$

Along a common unbounded set of cutoffs J for that R, assume jointly

$$\delta_{J,R} \to 0, \qquad \delta_{J,R} n_{J,R} \to 0, \qquad \eta_J/\delta_{J,R} \to 0. \tag{5.7}$$

Then there are $\varepsilon > 0, d_0$ such that $a_0 \leq \varepsilon L$ and $d \geq d_0$ imply $\Delta \geq \varepsilon L$ for this class.

Proof (not formalized). Put $w = \lceil \delta_{J,R} L \rceil$. At every grid shift, the blocks meeting any one covering interval have total capacity at most $3w$. Thus the raw revelation in Proposition 5.1 costs at most $3w\mathcal{N}_{\rho,J,R}(\delta_{J,R})$. Its dimension limit replaces the right side of Eq. (5.5) by

$$2\delta_{J,R} + 3\delta_{J,R} n_{J,R} + A_R \eta_J/\delta_{J,R}.$$

All three terms vanish along the cutoffs in Eq. (5.7). The proof of Theorem 5.2 now gives the same contradiction. □

Corollary 5.5 (Sublinear raw covering exponents)**.** Not formalized; a complete proof is given below.
The metric criterion holds in either of the following classes, with constants uniform in dimension and cutoff and permitted to depend on each fixed R in an unbounded set.

(a) Without a size envelope, $\mathbb{E}_\rho \mathcal{N}_{\rho,J,R}(\delta) \leq H_R \delta^{-s_R}$ for all $J \geq R$ and $0 < \delta < \delta_{0,R}$, where $H_R, \delta_{0,R} > 0$ and $0 \leq s_R < 1$.

(b) Under $|G_i| \leq HL/i^p$ with fixed $p > 1$, $\mathbb{E}_\rho \mathcal{N}_{\rho,J,R}(\delta) \leq H_R J^{u_R} \delta^{-s_R}$ in the same range, where $0 \leq s_R < 1$ and $0 \leq u_R < p(1 - s_R)$.

Proof (not formalized). In (a) take $\eta_J = c/(J+1)$ and $\delta_{J,R} = (J+1)^{-1/2}$. In (b) choose a fixed exponent $u_R/(1-s_R) < v_R < p$ and set $\delta_{J,R} = (J+1)^{-v_R}$, with $\eta_J = \min\{c/(J+1), H/(J+1)^p\}$. The products δn are respectively bounded by constants times $(J+1)^{-(1-s_R)/2}$ and $J^{u_R - v_R(1-s_R)}$; the residual ratios η_J/δ also tend to zero. Eventually $\delta < \delta_{0,R}$. Apply Proposition 5.4. □

These conditions concern raw-coordinate support shared by all outputs. They can be more economical than counting separate temporal occurrences. Supplying them for a still-open native class remains a geometric problem; the hypotheses are not consequences of a low entropy estimate alone.

6 Why complete continuation matters

The source-cover criteria require bounds on complete raw recovery. Neither disjoint exceptional spans nor small variation of raw reads transported through the exceptional maps supplies such a bound by itself. We give one native geometry that makes this distinction explicit. It has a quadratic fractional cover at one root and phase, even though all exceptional-only transported profiles vanish on the nondistinguished core families. The example does not satisfy a fixed inverse-square family-size envelope and is not a counterexample to an averaged cover criterion.

We use the strictly clipped shadows of Definition 2.4, with deleted families acting as the identity. Their ordinary orbit includes the seed and is not truncated at other paid vertices. It is therefore distinct from the actual dependency cone after paid lower vertices have been removed.

For a retained core $\mathcal{K}$, let $B_{\mathcal{K}}(u)$ be the raw labels of its ordinary orbit from time u. For a nonempty raw set S, define its arbitrary-seed fractional core-cover number by

$$\kappa_{\mathcal{K}}^{\mathrm{all}}(S) = \inf\left\{ \sum_{u=0}^{T-1} w_u : w_u \geq 0, \quad \sum_{u:\, a \in B_{\mathcal{K}}(u)} w_u \geq 1 \text{ for every } a \in S \right\}. \tag{6.1}$$

Allowing every computation time as a seed can only make covering easier than requiring the seeds to lie in the original joint orbit.

Proposition 6.1 (Serial exceptional spans can require a quadratic cover)**.** Not formalized; a complete proof is given below.

For every integer $M \geq 4$ and $b \geq 1$, there is a prescribed native geometry with

$$d = 16M, \qquad L = d^2 b,$$

eight work heads, unit moves or pauses, at most three entries per work cell, fewer than $13L$ time vertices, and L distinct increasing output times, with the following properties. In particular the construction allows the logarithmic entry width $b = b(d)$.

The complete actual return families consist of a core $\mathcal{K}_0$ whose full spans partition into two disjoint-span classes, and two exceptional families F, G with disjoint full spans. There are no residual families. A period-M deletion of complete core families, together with payment of every selected outer lower endpoint, produces a retained core $\mathcal{K}$, a paid set Γ, and an output root t such that, writing $S(t)$ for its complete ordinary raw orbit and $Q = (M-1)^2$,

$$\kappa_{\mathcal{K}}^{\mathrm{all}}(S(t)) = Q, \qquad \mathcal{A}(S(t)) \leq \frac{1}{2Q} + \frac{1}{2L}. \tag{6.2}$$

The same cover equality holds if covering seeds must lie in the joint orbit of t. The paid physical dependency cone at t contains Q raw coordinates with circular gaps at most $\lceil L/Q \rceil$.

There is one core family H such that, on the entire full span of every other original core family, $x \circ w$ is identically zero for every finite word w in the two exceptional shadows, including the empty word. Here x is the raw input-head position. Thus all these exceptional-only variation tails vanish, despite the quadratic cover.

Proof (not formalized). Put

$$m = M, \quad n = 8m, \quad h = 4nb, \quad L = nh = 4n^2 b, \quad A = 4L, \quad E = 11L, \quad T = 12L + h + 1.$$

All heads begin at zero. Four heads form two copies of the following bank construction, beginning at times A and E. At time $D + 2ih$, $0 \leq i < n/2$, head $i \bmod 2$ moves up h, down h, and up h from its current base, each leg at unit speed; it waits until its next bank. Before and after these movements it remains parked. The base increases by h after each bank on that head. Direct inspection of the successive visits gives exactly the families

$$\begin{aligned} C_j^D : \quad & \text{span } (D + jh, D + (j+2)h], \quad \text{owner } D + (j+1)h, \\ & \text{hops } (D + (j+1)h - u, D + (j+1)h + u), \quad 1 \leq u \leq h, \end{aligned} \tag{6.3}$$

for $D \in \{A, E\}$ and $0 \leq j < n$. The two families in each bank come from its upper and lower turn respectively. Its interior cells are entered on its three legs; a later bank uses a fresh spatial interval. The common boundary cell also has at most three entries. Parking adds only adjacent previous

visits. Thus Eq. (6.3) is the complete nonadjacent family list on those four heads. Even and odd j give two disjoint-span classes.

A fifth head supplies F: it waits through $7L$, rises to L by $8L$, holds through $11L$, and descends to zero by $12L$. Its one complete family has

$$\text{span } (7L, 12L], \quad \text{owner } 8L, \quad \text{hops } (7L + a, 12L - a), \quad 0 \le a < L.$$

A sixth head supplies the core family H: it rises to L by time L, holds through $2L$, and descends to zero by $3L$. Its family is

$$\text{span } (0, 3L], \quad \text{owner } L, \quad \text{hops } (a, 3L - a), \quad 0 \le a < L.$$

There is no later ascent on either head, hence no additional lower-turn family. Their plateaux contribute adjacent visits.

The eighth head will supply a core bridge B. Set $i = 4m - 1$ and

$$r_k = 8L - (i - k)h - h/2, \qquad 0 \le k < m.$$

This head waits through $5L$, rises to L by $6L$, descends at unit speed to height m by $7L - m$, and then waits. It descends by one at each time $r_0, \dots, r_{m-1}$ and parks at zero. These times increase, and $r_0 > 7L - m$, so the schedule is legal. Its complete family has owner $6L$, span $(5L, r_{m-1}]$, and hops

$$(5L + a, 7L - a) \quad (m \le a < L), \qquad (5L + m - 1 - k, r_k) \quad (0 \le k < m).$$

In particular B has L actual hops. Among bank spans it overlaps only C^A_{n-1}: the preceding span ends exactly at $5L$. Assign B to the even class and H to that class as well. The entire core still has two disjoint-span classes; the early overlap graph is the path $C^A_0, \dots, C^A_{n-1}, B$.

The input head follows

$$x(t) = \begin{cases} t, & 0 \le t \le L - 1, \\ 2L - 2 - t, & L - 1 \le t \le 2L - 2, \\ 0, & t \ge 2L - 2. \end{cases} \tag{6.4}$$

All raw coordinates have been visited before the prescribed output times $E, E + 1, \dots, E + L - 1$.

We next specify the seventh head and the paid phase. On the early overlap path give C^A_j rank $\min\{\max\{j, 1\}, n - 1\}$ and B rank $n - 1$. On the late path give C^E_j rank $\min\{\max\{j, 1\}, n - 2\}$; give H rank one. Delete rank residue zero modulo m. This deletes precisely $C^D_m, C^D_{2m}, \dots, C^D_{7m}$ in each bank component and retains H, B. These are also the path-spine ranks with the endpoint leaves assigned their adjacent spine rank.

Choose

$$t = E + (4m - 1)h + h/2.$$

Its retained late-core orbit consists of $v_k = E + (4m - 1 - k)h + h/2$, $0 \le k \le m - 2$. Indeed each reflection reduces the window index by one, until the next required family is C^E_{3m}, which was deleted. The actual F hop from v_k lands at r_k. The actual B hop then lands at $5L + a_k$, where $a_k = m - 1 - k$. The last early bank family sends this point below the bridge's outer lower endpoint:

$$5L + a_k \longmapsto 5L - a_k < 5L.$$

Keeping the complete older continuation gives the $Q = (m - 1)^2$ times

$$z_{k,\ell} = A + (n - \ell)h + \begin{cases} a_k, & \ell \text{ even}, \\ h - a_k, & \ell \text{ odd}, \end{cases} \quad 0 \le k \le m - 2, \quad 1 \le \ell \le m - 1. \tag{6.5}$$

They are distinct: different ℓ give different windows, and within a window the offsets a_k are distinct. They are all strictly interior to their windows, their maximum is $5L-1$, and their lowest window is $7m+1$. The next older family C^A_{7m} was deleted, so these are the complete older bank continuations.

Order these times as $z_1 < \cdots < z_Q$ and set

$$u_j = 2L + \lfloor jL/Q \rfloor, \qquad 0 \le j < Q.$$

This ordering defines the prescribed geometry; it is not an uncharged sorting operation in a claimed algorithm. The seventh head waits through $u_0 = 2L$. It leaves cell j at time $u_j + 1$, reaches cell $j+1$, and waits until its next specified move. Its first peak is at $u_{Q-1}+1 < 3L$. It holds at height Q until $z_1 - 1$, then descends by one at each z_i and parks. Since both sequences are strictly increasing, all moves are unit moves or pauses. Its only family G has

$$\text{span } (2L, z_Q] = (2L, 5L-1], \quad \text{owner } u_{Q-1}+1, \quad \text{hops } (u_{Q-i}, z_i), \quad 1 \le i \le Q.$$

The lower endpoints use the last outward visits, including all waits. The full spans of G and F are disjoint and can share a third disjoint-span class. We have now listed all $2n+4$ complete families; the new heads have no later ascent and no omitted opposite-side family. All work cells have at most three entries.

Pay every actual lower endpoint of the fourteen deleted bank families and every outer lower endpoint in the full selected family set. The first set has $14h$ elements. The second has $2n+4$ elements, of which fourteen already belong to the first. Therefore

$$|\Gamma| = 14h + 2n - 10. \tag{6.6}$$

Every displayed path to a z_i avoids these payments: all bank offsets are interior, all its windows avoid the deleted lower windows, and the F and B hops avoid their outer lowers. For $1 \le j < Q$ the actual G and H hops then give

$$z_{Q-j} \longmapsto u_j \longmapsto L - \lfloor jL/Q \rfloor.$$

These vertices also avoid Γ. The last G hop lands at the paid outer lower $u_0 = 2L$ and is discarded in the physical cone. Together with the root's direct read zero, the surviving paths prove that the physical cone contains

$$R = \{0\} \cup \{L - \lfloor jL/Q \rfloor : 1 \le j < Q\}. \tag{6.7}$$

All state predecessors, other work predecessors and direct reads are still retained; they can only add sources to this witnessed subset. Its Q circular gaps are at most $\lceil L/Q \rceil$. Inserting points only decreases the sum of squared gaps, so

$$\mathcal{A}(S(t)) \le \mathcal{A}(R) \le \frac{\lceil L/Q \rceil}{2L} \le \frac{1}{2Q} + \frac{1}{2L}.$$

For the exact cover assertion, the ordinary joint time orbit consists of six groups: the $m-1$ times v_k, their $m-1$ images r_k, the $m-1$ images $5L + a_k$, the Q times z_i, their Q strict G images, and the Q subsequent H images. The calculations above show that each is reached. They also show that the union is closed under all retained maps: within a bank the only change is the next listed reflection; B acts on the r_k; F acts on the v_k; G acts on the z_i; and H acts on the G images. Other maps fix these points. At the last G return the ordinary clipping rule replaces $2L$ by $2L+1$, and H then sends it to $L-1$. Thus

$$S(t) = \{0, L-1\} \cup \{L - \lfloor jL/Q \rfloor : 1 \le j < Q\}. \tag{6.8}$$

The extra label $L-1$ here is an ordinary clipped source; its displayed clipped path is not asserted to survive the physical payment.

Give each of the Q nonzero labels in Eq. (6.8) weight one. A complete retained core orbit from any computation time contains at most one of them. To see this, after time $3L$ every core continuation stays after $3L$, where $x = 0$. Before time L the core orbit is a singleton. Between L and $2L$, the only possible extra core image is H's held owner L, whose raw value is $L-2$; this is not a target because $L/Q > 256$. Between $2L$ and $3L$, the root reads zero and the sole nonidentity core map H adds at most one raw value. The endpoints obey the same description. Summing the fractional covering constraints for the Q target labels therefore gives $\sum_u w_u \geq Q$.

Conversely, use the Q seeds

$$2L+1, \quad u_1, \ldots, u_{Q-1}.$$

Each lies in the joint orbit and has core raw set consisting of zero and one different nonzero target in Eq. (6.8). Weight one on each seed covers $S(t)$. This proves both asserted cover equalities.

Finally, x is zero throughout $[2L, T-1]$. Both exceptional shadows preserve that interval: their nonidentity images stay strictly above their outer lowers, which are $2L$ and $7L$, and elsewhere they are the identity. Every original core span except H lies in that interval. Hence $x \circ w$ vanishes on each such full span for every word w of exceptional shadows. This proves the profile assertion and completes the construction. □

The obstruction in this example is the continuation after the bridge exit. Many distinct intermediate times remain distinct through the older bank, and the final core return H converts them into separated raw coordinates. Looking only at raw reads immediately after an exceptional hop misses this last recovery step. In particular, Proposition 6.1 rules out a universal pointwise $O(M)$ fractional core-cover bound on the stated serial and zero-tail premises. It also rules out a uniform pointwise lower bound $\mathcal{A}(S(t)) \geq c/(M+1)$ with $c > 0$: along this family, $(M+1)\mathcal{A}(S(t)) \to 0$.

There are two essential restrictions on that conclusion. First, the example ties $d = 16M$ to the growing phase parameter and supplies one root and one phase. It does not disprove a phase or column average whose parameters are fixed in the order required by the uniform source-cover theorem. Second, selecting the four complete families $\{H, B, F, G\}$ leaves maximum family size $h = L/n = o(L)$. Thus Corollary 4.5 already excludes near-full correctness on this sequence of geometries; it does not provide a new hard native class. Its $2n$ equal bank families also violate every common fixed inverse-square envelope: the nth largest core family has at least $h = L/n$ hops, so a bound $h_j \leq CL/j^2$ would force $C \geq n$. These are prescribed legal geometries and source sets, not a transposition algorithm or a proof that every reachable source carries necessary information. An affordable cut, an averaged cover bound, or a different complete recovery certificate may still handle them.

7 Movement required by a specified recovery interface

A source cover must satisfy two different requirements: it must contain enough information to recover the output, and its source positions must have an affordable joint trajectory. This section separates those requirements on a familiar coded-caching example. The code itself is known. The movement bounds concern its specified raw-query interface; they do not exclude every decoder for that code.

7.1 A finite bound independent of query order

Write dist_L for distance on the circle of circumference L. Fix $N \geq 1$ equally spaced centres $c_i = (i + \frac{1}{2})L/N$, $0 \leq i < N$. A query type is a k-element subset of these centres, where $1 \leq k \leq N$.

Theorem 7.1 (Theorem M: movement to cover distinct query types)**.** Not formalized; a complete proof is given below.

Let $S_1, \ldots, S_J$ be a nonempty sequence of query types, with F distinct types. For $1 \leq \ell \leq m$, let $y_\ell(j)$ be labelled source positions on the circle, where $m \geq 1$. Suppose every centre in S_j is within distance $\rho > 0$ of some $y_\ell(j)$. Put

$$V = \sum_{\ell=1}^{m} \sum_{j=1}^{J-1} \mathrm{dist}_L(y_\ell(j+1), y_\ell(j)), \qquad v = V/L, \qquad \lambda = \rho/L,$$

$$s = \min\left\{N,\ m\left(\left\lfloor \frac{5\lambda N}{2} \right\rfloor + 1\right)\right\}, \qquad B = \binom{s}{k},$$

with $\binom{s}{k} = 0$ for $s < k$. Then

$$F \leq (1 + 4v/\lambda)B. \tag{7.1}$$

In particular $B > 0$, and

$$v \geq \frac{\lambda}{4}\left(\frac{F}{B} - 1\right). \tag{7.2}$$

The bounds hold for every ordering of the queries and every matching of source positions between successive queries. They also hold when V is measured by ordinary distance on the input line.

Proof (not formalized). Set $\delta = \rho/4$. Greedily partition the sequence into consecutive stretches whose internal summed movement is at most δ: end a stretch immediately before the next edge would exceed that threshold. Charge a nonfinal stretch its internal edges and the edge immediately after it. These charged edge sets are disjoint, and each has total weight greater than δ. Thus there are at most $1 + V/\delta$ stretches, including the last.

Within a stretch, each labelled position stays within distance δ of its initial position. Every queried centre is therefore within $5\rho/4$ of one of the m initial positions. A circular ball of that radius contains at most

$$\min\left\{N, \left\lfloor \frac{5\rho N}{2L} \right\rfloor + 1\right\}$$

of the equally spaced centres. Indeed, if the ball is a proper arc, its length is $5\rho/2$, and consecutive centres are L/N apart; if it is the whole circle, use N. The union of the balls contains at most s centres and hence at most B query types. Summing over stretches proves Eq. (7.1); rearranging proves Eq. (7.2). A nonempty sequence rules out $B = 0$.

No step used the ordering of the types. Any chosen matching of the m positions between successive queries defines coherent labelled paths; apply the same argument to those paths. Finally, circle distance never exceeds line distance, so replacing circle variation by line variation preserves the bound. □

For a tape construction, this theorem charges the movement of the entire source bank. Relabelling the bank slots does not remove that charge.

Corollary 7.2 (Queries spread over a column interval)**.** Not formalized; a complete proof is given below. *Let $I_1, \dots, I_J$ be disjoint, chronologically ordered intervals in a finite discrete timeline. Let m labelled source trajectories have total circle variation V over that entire timeline, including the gaps between intervals. Assign a query type S_j to each interval, and suppose each of the F distinct types occurs at least $r \geq 1$ times. For every $c_i \in S_j$, suppose there is a raw point $a_{i,j}$ with*

$$\mathrm{dist}_L(a_{i,j}, c_i) \leq w$$

that is within distance ρ of a source position at some time in I_j. The position and time may depend on i and j. With $v = V/L$, $\lambda = \rho/L$, and

$$s' = \min\{N,\ m(\lfloor 3\lambda N + 2wN/L \rfloor + 1)\}, \qquad B' = \binom{s'}{k},$$

one has

$$F - \frac{4v}{\lambda r} \leq (1 + 4v/\lambda)B', \tag{7.3}$$

$$v \geq \frac{\lambda}{4} \frac{F - B'}{B' + 1/r}. \tag{7.4}$$

Proof (not formalized). Call an interval bad if its internal summed movement exceeds $\rho/4$. There are at most $4V/\rho = 4v/\lambda$ bad intervals. Removing them can remove every occurrence of at most $4v/(\lambda r)$ types.

Freeze the positions at the start of each remaining interval. A required centre is within distance $\rho + \rho/4 + w$ of one of these frozen positions. The sequence of frozen positions has total variation at most V, by the triangle inequality on each labelled trajectory. Partition that sequence into stretches of movement at most $\rho/4$ as in Theorem 7.1. Within each stretch, every required centre lies within distance $3\rho/2 + w$ of one of its initial positions. There are at most s' such centres and at most B' types. This proves Eq. (7.3), including when no good intervals remain. Collecting the terms involving v gives Eq. (7.4). □

Thus queries made at different times during an output column do not permit us to freeze their positions for free. Their internal movement and the possible loss of query types both appear in the bound.

7.2 A coded-caching calibration

We use the transpose of the subset placement-delivery array in [9, Section IV, Theorem 2]. Repeating its rows and columns gives the square matrix below. We include the elementary decoding and entropy calculation to specify exactly which interface is being tested.

Fix integers $N > k \geq 2$ and $h \geq 1$, and put

$$F = \binom{N}{k}, \qquad d = NFh, \qquad L = d^2 b, \qquad W = db, \qquad a = L/N,$$

where $b \geq 1$ is an integer. The asymptotic application permits $b = b(d)$. Index the rows by $(i, u) \in \{0, \dots, N-1\} \times \{0, \dots, Fh-1\}$, grouping each fixed i into one contiguous bin. Index columns by (S, z), where $|S| = k$ and $0 \leq z < Nh$. A column decoder may observe all raw rows in the bins $i \in S$, together with one common helper G. The helper contains the following b-bit parities, for every $(k+1)$-element set T and every u, z:

$$Z_{T,u,z} = \bigoplus_{i \in T} X_{(i,u),(T\setminus\{i\},z)}. \tag{7.5}$$

Proposition 7.3 (Exact helper and gap-area calibration)**.** Not formalized; a complete proof is given below.

The specified interface recovers every column on the full input cube. For a uniform input, the helper is uniform on g bits, where

$$g = H(G) = L\frac{N-k}{N(k+1)}. \tag{7.6}$$

Every helper fibre has size 2^{L-g}. Every entry with $i \in S$ is absent from all helper parities and is independent of G and all other input entries.

For $P_S = \{ia : i \in S\}$ on the raw circle,

$$\frac{1}{\binom{N}{k}} \sum_{|S|=k} \mathcal{A}(P_S) = \frac{2N-k+1}{2N(k+1)} = \frac{g}{L} + \frac{1}{2N}. \tag{7.7}$$

Consequently, for fixed k and $N \to \infty$, the queried row fraction k/N tends to zero, whereas both g/L and the mean point-gap area tend to $1/(k+1)$.

Proof (not formalized). If $i \in S$, the desired entry is read directly. If $i \notin S$, take $T = S \cup \{i\}$ in Eq. (7.5). All the other terms have row-bin index in S, so the queried rows and that parity determine the entry. Every nonqueried entry belongs to exactly one parity, specified by $T = S \cup \{i\}$ and its u, z. The parity classes are disjoint, so their bits are independent and uniform. Counting them gives

$$g = b\binom{N}{k+1}(Fh)(Nh) = L\frac{N-k}{N(k+1)}.$$

The parity map is surjective, with all fibres of size 2^{L-g}. Queried entries occur in none of these parity classes, proving the independence assertion.

For the gap formula, choose S uniformly and mark one of its elements uniformly. Starting at the marked element, the clockwise integer gaps $D_1, \ldots, D_k$ are uniform among positive compositions of N. Indeed, each composition and starting point specifies one marked set, and $N\binom{N-1}{k-1} = k\binom{N}{k}$. Put $E_i = D_i - 1$ and $n = N - k$. Then $\sum_i E_i = n$. Stars and bars gives

$$\mathbb{E}\sum_i E_i(E_i-1) = \frac{2k\binom{n+k-1}{k+1}}{\binom{n+k-1}{k-1}} = \frac{2n(n-1)}{k+1},$$

with both sides zero when $n < 2$. The numerator follows, for example, by taking the coefficient of z^n in $2kz^2(1-z)^{-k-2}$. Expanding $\sum_i (E_i+1)^2$ now yields

$$\mathbb{E}\sum_i D_i^2 = \frac{N(2N-k+1)}{k+1}.$$

The geometric gaps are aD_i, so division by $2L^2$ proves Eq. (7.7). The limits follow from the displayed formulas. □

This example prevents an information-only inference from sparse queried rows to a deficit exceeding the helper cost. The exact recovery code exists, and the normalized gap area approaches the same value as that cost. The remaining issue is its geometric realization.

Corollary 7.4 (The dimension limit precedes the radius limit)**.** Not formalized; a complete proof is given below.

Fix $k \geq 2$, $m \geq 1$ and $h \geq 1$, and take the preceding interfaces with $N \to \infty$. For each N, allow arbitrary column order and arbitrary coherent source matching. Let v_N be the total circle variation divided by L.

If every queried bin centre is covered at the column representative within radius $\rho = \lambda L$, then, for every fixed $0 < \lambda < 2/(5m)$,

$$\liminf_{N\to\infty} v_N \geq \frac{\lambda}{4}\left[\left(\frac{5m\lambda}{2}\right)^{-k} - 1\right]. \tag{7.8}$$

If instead one required raw point within W of each queried bin centre is covered within radius λL at some time during its column interval, then, for every fixed $0 < \lambda < 1/(3m)$,

$$\liminf_{N\to\infty} v_N \geq \frac{\lambda}{4}[(3m\lambda)^{-k} - 1]. \tag{7.9}$$

Both right-hand sides tend to infinity as $\lambda \downarrow 0$. Thus no fixed variation constant represents this interface with a fixed number of source trajectories at every sufficiently small fixed radius ratio.

Proof (not formalized). For the first assertion, $s/N \to 5m\lambda/2 < 1$ in Theorem 7.1. Since k is fixed,

$$\frac{\binom{s}{k}}{\binom{N}{k}} \longrightarrow (5m\lambda/2)^k.$$

Apply Eq. (7.2). For the second assertion, each type occurs $r = Nh$ times and $w = W = L/d$. Hence $2wN/L = 2/(Fh) \to 0$, $s'/N \to 3m\lambda < 1$, and $\binom{s'}{k}/\binom{N}{k} \to (3m\lambda)^k$. Divide the numerator and denominator of Eq. (7.4) by $\binom{N}{k}$ to obtain Eq. (7.9). Each lower bound is a positive constant times λ^{1-k} minus $\lambda/4$, which proves the last assertion. □

The necessary-point premise has a precise information-theoretic use. Choose, for each queried bin, one bit of an own-column entry in a middle row. Its raw coordinate is within W of the bin centre. That bit is independent of the specified helper and every other raw bit, so a decoder on the full cube cannot omit it from its raw dependency set. Any geometric representation of that dependency set must cover it. When those raw observations are represented during the column interval, Corollary 7.4 applies.

This necessity assertion does not survive every change of helper. Revealing all entries with $i \in S$ costs exactly $kL/N = o(L)$ extra bits at fixed k and removes these particular witnesses. That payment does not show that the remaining recovery is cheap; it shows why the movement result is about the stated interface. A native source-bank argument may reveal additional threshold or residual information, so its required queries must be established after those exact disclosures. The radius ratio is also fixed before the dimension limit; the conclusion does not interchange these limits by setting $\lambda = 1/N$.

8 Causally autonomous column groups

The source-cover results restrict the geometry of complete dependence. This section gives a complementary restriction on the functions computed along that geometry. If every local update obeys a capacity-conserving decomposition into fixed narrow groups of consecutive columns after one helper is fixed, then a linear-length computation must disclose almost the entire input through that helper. Neither the return hypotheses nor final column purity alone supplies this causal decomposition.

8.1 The causal hypothesis and the routing convention

Here $b \geq 1$ is any integer, $W = db$, and $L = d^2 b$. Consider a directed acyclic network with nodes $0, \ldots, n-1$, all edges directed from smaller to larger nodes, and all memory edges $(v, v+1)$. Edge e has a fixed nonnegative capacity c_e, measured in bits. Each input row i has a specified set A_i of attachment nodes; at a node in A_i the entire raw row is visible. Each raw bit $(i, j, t) \in [d]^2 \times [b]$ has a specified output node $o(i, j, t)$. These nodes respect transposed output order: the bits of column j are output before those of column $j+1$, although several columns can share a node. These are the row-source and output conventions of the block-network model in [3, section "Codes on time–space diagrams"].

Let $X \in \{0,1\}^L$ have any probability law supported on correctly transposed inputs, and let Z be a deterministic function of X. Partition the columns into nonempty consecutive intervals $J_1, \ldots, J_h$, in increasing order. Write $g_a = |J_a|$, $g = \max_a g_a$, and $X_a = X|_{[d] \times J_a \times [b]}$. All entropies and logarithms in this section use base 2.

Definition 8.1 (Causal group typing)**.** The computation has a *capacity-conserving causal group typing conditional on Z* if the following holds for each attained z. For each edge e there is one tuple of fields of each type $a \in [h]$, taking values in an alphabet of size at most $2^{c_{e,a}(z)}$, where

$$c_{e,a}(z) \geq 0, \qquad \sum_{a=1}^{h} c_{e,a}(z) \leq c_e.$$

Every outgoing field of type a is a deterministic function only of incoming type-a fields, the raw bits of X_a in rows attached at that node, and z. Each output bit in group a is obtained by a function with these same permitted arguments. The field alphabets, allocations and local functions are fixed on the whole fibre $Z = z$, and their evaluation agrees there with the computation and its outputs.

All input-dependent control used by a local function must consequently be carried by its permitted fields or determined by z. A shared symbol cannot be reused as a full-capacity field for several groups without paying the sum of those capacities. The local functions may be nonlinear and need only match the original computation on the fibre; they can be extended arbitrarily to unattained local tuples. Merely asserting that an edge's final value depends on one group does not establish this local causal factorization.

A *partial routing* below uses the underlying undirected network. For each raw bit it assigns mass in $[0, 1]$ to walks from nodes in its row's attachment set to its own output node. Every edge traversal, in either direction and with multiplicity, consumes capacity. Bits of unattached rows receive zero mass. This is an analytic fractional routing; it is neither a causal execution nor a routing at one common rate for all bits.

For each group let α_a and ω_a be its first and last output nodes, and put

$$I_a = \{(v, v+1) : \alpha_a \leq v < \omega_a\}, \qquad W_a = dbg_a.$$

The edge intervals I_a are pairwise disjoint, since the groups are consecutive. They may share endpoint nodes. Define the paid volume

$$V = \sum_e c_e + 2 \sum_{a=1}^{h} W_a(\omega_a - \alpha_a) \leq \sum_e c_e + 2gdb(n-1). \tag{8.1}$$

The factor 2 pays both collection at ω_a and return to each bit's original output node.

8.2 From autonomous recovery to partial routing

Theorem 8.2 (Theorem Q: paid routing from causal groups)**.** Not formalized; a complete proof is given below.

Assume Definition 8.1. For every attained z, let

$$\mathcal{A}_a(z) = \{x|_{[d]\times J_a\times[b]} : x \in \operatorname{supp}(X \mid Z = z)\}, \qquad R_{\sup} = \sum_z \Pr[Z = z] \sum_{a=1}^{h} \log_2 |\mathcal{A}_a(z)|.$$

There is a partial routing of total mass $R_{\sup}$, with capacity on edge e at most

$$c'_e = c_e + 2 \sum_{a:e\in I_a} W_a. \tag{8.2}$$

For an integer $\ell \geq 1$, let S_ℓ be the number of raw bits whose row has an attachment and whose distance from its nearest row attachment to its original output node is less than ℓ. Then

$$\sum_{a=1}^{h} H(X_a \mid Z) \leq R_{\sup} \leq S_\ell + \frac{V}{\ell}. \tag{8.3}$$

In particular, writing $\Delta = L - H(X)$ and

$$\mathrm{TC}(X_1; \ldots; X_h \mid Z) = \sum_a H(X_a \mid Z) - H(X \mid Z),$$

one has

$$\Delta + H(Z) \geq L + \mathrm{TC}(X_1; \ldots; X_h \mid Z) - S_\ell - \frac{V}{\ell}. \tag{8.4}$$

No independence of the input rows, groups, or helper fibres is assumed.

Proof (not formalized). Fix a, z. Causal induction shows that evaluating all fields of type a depends only on X_a and z. Indeed, this is true of the locally visible raw data, and it is preserved at each node by the local typing rule. Thus a projected input $y \in \mathcal{A}_a(z)$ defines a code recovering every bit of y, independently of which full input extending y was used to define it. This argument evaluates one projection at a time; it makes no product assumption about the conditional support.

Add a forward collector of capacity W_a along each edge of I_a. It carries the previously output bits of group a, in their fixed order. At node ω_a this prefix together with the outputs at that node determines all of y. These collector fields still obey group autonomy. Their total added capacity is $W_a(\omega_a - \alpha_a)$, including zero when both endpoints coincide.

Add a supersource s and a source vertex u_i for each row. Give (s, u_i) capacity bg_a, and give every arc from u_i to a node of A_i capacity $L + 1$. The sink is ω_a. Consider any s–ω_a cut. If it crosses one of the latter attachment arcs, its capacity exceeds $L \geq \log_2 |\mathcal{A}_a(z)|$. Otherwise let U be its sink-side physical nodes and let B be its sink-side row vertices. Every row attached in U belongs to B.

For an input $y \in \mathcal{A}_a(z)$, record its bg_a bits in each row of B and all type-a fields and collector values entering U. This tuple, with z fixed, determines the computation inside U in node order. It supplies every raw argument visible there and every parent value entering from outside, including any edge that re-enters U after an earlier departure. It therefore determines y at the collector sink. The tuple map is injective, giving

$$\log_2 |\mathcal{A}_a(z)| \leq bg_a|B| + \text{capacity entering } U.$$

The right side is the cut capacity. The directed max-flow/min-cut theorem, whose original undirected form is due to Ford and Fulkerson [7], supplies a flow of value $\log_2 |\mathcal{A}_a(z)|$, truncated to that value if necessary. For completeness, the directed version with real capacities follows by maximizing flow value over the compact feasible-flow polytope. A path of positive residual capacities to the sink would increase that maximum; if no such path exists, the residual reachable set gives a cut of the same value. Row i supplies at most bg_a, and an unattached row supplies zero.

Decompose the flow into source-to-sink paths and remove their artificial source and attachment arcs. Split the path mass from each row among its bg_a original group bits, assigning at most one unit to each bit. This is possible because all those bits have the same permitted row attachments. It need not identify particular bits as entropy carriers. For each allocated bit append the memory path backwards from ω_a to that bit's original output node. Every such return stays in I_a and contributes at most the total group flow, hence at most W_a, to each edge of that interval. The forward collector and these returns together cost at most $2W_a$ per edge of I_a.

Sum over groups. On an original edge the group flows use at most $\sum_a c_{e,a}(z) \le c_e$. The collector and return loads obey Eq. (8.2). Average these routings over z. Their common capacity constraints and per-bit mass bounds are linear, so the average is feasible and has total mass $R_{\sup}$. Also $H(X_a \mid Z = z) \le \log_2 |\mathcal{A}_a(z)|$, proving the first inequality in Eq. (8.3).

At most S_ℓ routed mass belongs to short-distance bits, since each bit has mass at most one. Every unit of the remaining mass traverses at least ℓ physical edges. Consequently $\ell(R_{\sup} - S_\ell) \le V$, proving the second inequality. Unattached rows contribute no routed mass and are not counted as zero-distance sources. Finally, because Z is determined by X,

$$\sum_a H(X_a \mid Z) = L - \Delta - H(Z) + \mathrm{TC}(X_1; \dots; X_h \mid Z).$$

Substitution proves Eq. (8.4). □

The theorem supplies an explicit conversion from a restricted causal code to partial routing. Its extra communication is fully charged. It does not strengthen the general routing obstruction, and it does not convert arbitrary causal codes to routings.

8.3 Application to linear-length tape diagrams

For clarity we state the precise short-distance estimate needed for partial, possibly unequal, commodity masses. The summed-distance bound alone would not suffice.

Lemma 8.3 (Attached short-distance count)**.** Lean: `SourceCovers.attachedShortBits`, `SourceCovers.card_attachedShortBits_le_local`, `SourceCovers.card_attachedShortBits_le_profile`, `SourceCovers.card_attachedShortBits_le`, `SourceCovers.attachedShortBits_zero_radius`, `SourceCovers.attachedShortBits_zero_block` [`SourceCovers/NativeShortCount.lean`]

Let a k-work-tape diagram have block length $r \ge 0$, n block nodes, and raw length $L = d^2 b$. Use its original row attachments and transposed-bit output nodes. Write $R(a)$ for the rows attached at node a, $B_t(a)$ for its closed undirected radius-t ball, and $m_{i,s}$ for the number of bits of raw row i assigned to output node s. For every integer $\ell \ge 1$,

$$S_\ell \le \sum_{a<n} \sum_{i \in R(a)} \sum_{s \in B_{\ell-1}(a)} m_{i,s}. \tag{8.5}$$

This bound requires neither diagram validity nor an upper bound on r. If the diagram is valid for output length L and $r \le db$, then

$$S_\ell \le b \sum_{a<n} |R(a)|\,|B_{\ell-1}(a)|, \tag{8.6}$$

$$S_\ell \le \min\{L,\ (2n-1)_+ b(2k+3)^{\ell-1}\}. \tag{8.7}$$

Also $S_0 = 0$, and $S_\ell = 0$ for $r = 0$ under the library's zero-block convention. The Lean inequalities include these cases, interpreting natural subtraction $\ell - 1$ as $\max\{\ell-1, 0\}$ and $(2n-1)_+ = \max\{2n-1, 0\}$.

Outline of the formal proof. The formal definition excludes unattached rows, whose inherited default distance is zero. If an attachment exists, the block network is nonempty, its chosen output node lies in that network, and the memory spine connects it to the attachment. These facts require no validity hypothesis. The bridge lemma therefore identifies a short bit with a row attachment a whose radius-$(\ell-1)$ ball contains its output node. Taking the union over a, its attached rows, and its ball gives Eq. (8.5) by Mathlib's finite-union bound.

For a valid diagram and $1 \le r \le db$, a block's output positions lie in an interval of length at most db. Two distinct bits of one raw row with the same within-entry offset have transposed positions separated by a nonzero multiple of db. They cannot both lie in this interval. The within-entry offset thus injects the bits counted by $m_{i,s}$ into $\{0, \dots, b-1\}$, proving $m_{i,s} \le b$ and Eq. (8.6). This sharpens the first paper's $2b$ estimate, which separately counts the portions in two output columns [3, section "Codes on time–space diagrams"].

A block attaches at most two raw rows, and its underlying graph has degree at most $2k+2$. Reuse the first paper's `card_rowsAt_le` and `card_ball_le` from `Transpose/Tape/CECount.lean` to obtain $|R(a)| \le 2$ and $|B_t(a)| \le (2k+3)^t$. The first block can touch only row zero: the input head starts at zero and has fewer than $r \le db$ steps in that block. Consequently $\sum_{a<n} |R(a)| \le (2n-1)_+$. Together with $S_\ell \le L$, this gives Eq. (8.7). The zero-radius set is empty by its strict distance test; for zero block length there are no nodes or attachments. □

The profile bound retains actual row and ball counts for later use, without imposing a common routing rate on the counted bits. The sharper coefficient and first-block correction improve the finite estimate; they do not change the asymptotic obstruction below. No optimality of the graph-ball estimate or the union bound is claimed.

Corollary 8.4 (Narrow causal groups require an almost complete helper)**.** Not formalized; a complete proof is given below.

Fix $k \ge 0$, an integer $G \ge 1$, and $c > 0$. Along any sequence $d \to \infty$, let $b \ge 1$, $L = d^2 b$, and $r = \lceil db/G \rceil$. Suppose valid k-work-tape diagrams have length $T \le cL$, and their code capacities satisfy the block-profile volume bound

$$\sum_e c_e \le 4knrG, \qquad n = \lceil T/r \rceil.$$

Let X be any law on correctly transposed inputs and Z a raw-input-determined helper admitting causal group typing into fixed consecutive column groups of maximum size g. Then, with constants depending only on k, G, c,

$$\sum_a H(X_a \mid Z) = O\!\left(\frac{L}{\sqrt{d}} + \frac{(1+g)L}{\log d}\right). \tag{8.8}$$

In particular, if $g = o(\log d)$, then

$$\Delta + H(Z) \geq L - o(L).$$

Thus $\Delta = o(L)$ and $H(Z) = o(L)$ cannot both hold. The conclusion applies in particular to the series' standard width $b = b(d) = \max\{1, \lceil \log_2 d \rceil\}$.

Proof (not formalized). Since $r \geq db/G$, one has $n \leq cGd + 1 = O(d)$. Also $nr \leq T + r$ and $rG \leq db + G$, so $\sum_e c_e = O(L)$. Equation (8.1) therefore gives $V = O((1+g)L)$. For sufficiently large d, take $\ell = \lfloor \frac{1}{2} \log_{2k+3} d \rfloor \geq 1$. Then ℓ is of order $\log d$, and Lemma 8.3 gives $S_\ell = O(db\sqrt{d}) = O(L/\sqrt{d})$. Apply Theorem 8.2 to obtain Eq. (8.8). Finally, $L - \Delta - H(Z) = H(X \mid Z) \leq \sum_a H(X_a \mid Z)$. The right side is $o(L)$ when $g = o(\log d)$. □

At the standard width, the block-profile volume hypothesis is supplied by the Lean-verified `sum_cap_le` in [3]. The finite theorem also permits other profiles: a separate $O(L)$ total capacity allowance leaves the asymptotic conclusion unchanged.

For the construction program, the corollary excludes a completion that preserves one fixed narrow consecutive grouping at every local update after a sublinear-entropy joint helper is fixed. Arbitrarily many updates and arbitrary nonlinear operations inside each group remain allowed. It does not exclude final column purity, which every correct transposer already has, nor a succession of small groups that migrate and cumulatively mix all columns. For nonconsecutive groups the exact collector cost in Eq. (8.1) is still available, but its $2gdb(n-1)$ upper bound need not hold. Finally, none of the geometric hypotheses elsewhere in this paper is asserted to produce this causal typing with a cheap helper; $Z = X$ always gives a vacuous expensive one.

9 What remains for a transposition algorithm or impossibility proof

The main conclusion is a uniform deficit for specified full return geometries. It is stronger than a statement that one proposed code has failed: every deterministic local computation on any geometry in the stated class is covered, with no linearity assumption. Its scope is also explicit. A bounded full-span partition, a Q2 core, and a bounded number of exceptional families are additional geometric hypotheses. They have not been derived for an arbitrary linear-time transposer.

9.1 The nearest unresolved geometric case

For one exceptional family, Corollary 4.3 completes the deficit-versus-payment argument. With two exceptional families, a limiting inverse-square hop profile is compatible with Theorem 4.2. The next concrete problem is to determine whether the remaining complete continuation can satisfy all the necessary conditions simultaneously.

There are three precise ways to exclude a common native class in this case. First, bound the phase-and-column mean fractional cover by a common function $f(R) = o(R^2)$, uniformly in dimension and every finite first-J selection with $J \geq R$. Second, bound the mean number of globally maximal newer seeds, at each fixed protected scale R in a common unbounded set, by a function $g_R(J)$ with $\liminf_J g_R(J)/(J+1)^2 = 0$. Third, give the raw metric covers with the three joint limits in Eq. (5.7). These are respectively the criteria in Corollary 4.7, Theorem 5.2, and Proposition 5.4. They are alternatives; none is currently supplied for every admissible two-exception geometry with an inverse-square envelope.

A failed attempt should be checked against the exact hypothesis it tests. Proposition 6.1 defeats a universal pointwise linear cover estimate based only on serial exceptional spans and exceptional-only variation tails. It does not defeat a phase average, a common inverse-square envelope, or shared raw revelation across all outputs. It also belongs to a class already excluded by the finite skeleton corollary. Using it as evidence against all three uniform criteria would therefore draw a conclusion the example does not support.

Conversely, a candidate near-full sequence in the inverse-square class must meet the necessary reciprocal-cover, complete-observable movement, and global-continuation-density bounds of Theorem 4.6 and Corollaries 4.8 and 5.3, with all limits taken along the same joint profile. Producing such a geometry would test the sharpness of these certificates. It would not by itself supply a correct computation on the geometry.

9.2 What a constructive advance must compute

A recovery code becomes a transposition algorithm only after its encoding, updates, reads and output order have been implemented within the native time bound. The complete-source certificate charges one such implementation; it does not construct one from an abstract code. The coded-caching calibration makes this distinction concrete. Its algebraic decoder exists, but its specified raw-query interface cannot be represented by a fixed number of trajectories with uniformly bounded normalized movement at all small fixed radius ratios. Changing that interface or helper may evade the argument and must be analysed anew.

The causal-group result identifies another restricted shortcut that cannot work: after a cheap helper is fixed, a locally autonomous computation for fixed groups of $o(\log d)$ adjacent columns cannot retain nearly all the input information. A successful algorithm can avoid that premise by coupling groups during computation. The theorem does not price arbitrary such coupling. The constructive target is an explicit coupled update rule whose total native execution cost can be bounded, including the cost of changing where its encoded information is available. Final-output separation alone is not the forbidden causal typing.

9.3 What would decide (T)

An affirmative answer requires one fixed finite-tape, finite-alphabet machine, constants c, d_0, and a proof that it correctly transposes every input of every dimension $d \geq d_0$ in at most $cd^2b(d)$ steps. All preprocessing and recovery must be included. A family of dimension-dependent machines, a source relation, or a storage code without such an implementation does not meet this specification.

A negative answer must exclude every such fixed machine and constant. One possible route is a reduction from an arbitrary candidate to correctness sets and paid certificates with a strictly positive net deficit. Such a reduction must account for the cost of selecting a geometry in the adaptive case, and it must address geometries outside the span and core hypotheses used here. Alternatively, a different certificate may handle those geometries directly. The depth reductions of the fourth and eighth papers and the complete-recovery restrictions of the fifth provide complementary tools [2, 1, 5]; this paper does not prove that their hypotheses cover every remaining machine.

The immediate formalization task is correspondingly bounded: verify the exact finite certificates and their stated uniform consequences, without promoting a geometric criterion into a universal machine property. The specification in the following appendices separates those obligations from the unresolved question (T).

A Specification for subsequent formalization

Definitions 2.1, 2.2, 2.4, 3.1 and 4.1, Lemmas 2.3, 2.5, 2.6, 2.8, 3.2 to 3.4 and 8.3, Theorems 2.7, 2.9, 3.5, 4.2 and 4.6, Corollaries 3.6, 4.3 to 4.5, 4.7 and 4.8, and Propositions 3.7 and 5.1 are Lean-verified; their declarations and axiom commands accompany the paper. The remaining results retain complete written proofs. The index in Appendix B gives a stable label for every numbered result and definition; it is generated from the manuscript, rather than from a separate list of proposed theorems. Each result's status is explicit. The statements include the hypotheses to preserve when creating Lean declarations. Further formalizations must identify their declarations and disclose any change of scope before calling them authoritative.

A.1 Dependency order

1. *The model and complete-source engine.* Actual last-visit hops, full first-maximum families, the half-open span partition and family accounting are now formalized by reusing the eighth paper and finite incidence counting. Strict clipped shadows and their variation bounds, including successful words and disjoint packing, are also formalized, as are protected nonleaf phases and the exact packed bank, including closure across selected unpaid actual edges. Paid residual recovery is formalized for arbitrary locally determined records, with revealed-root reads and unpaid parents retained. The any-argument-path bound covers metric-valued paths; it is not restricted to the original input-head timeline.

2. *Finite deficit.* Circular gap area, one common cyclic ordering of all raw coordinates, sharp stripe discrepancy, and entropy recovery from a deterministic helper are formalized in Theorem 2.9. Its sharper finite bound retains maximum slot oscillations and improves the additive stripe error. The helper is charged by entropy, without inferring a support-cardinality bound. Minimum fractional core covers, interval rounding and the fractional raw-source gap bound are also formalized. The shared-root count sharpens its denominator to $2(2K+1)\kappa$; exact component incidences and union-edge variation are retained before this relaxation. Anchored erosion is now formalized with loss $e - e^2/2$, where $e = \min\{E/L, 1\}$, and the cover credit retains that saving. The full paid-cover assembly in Theorem 3.5 is now verified: the actual parent relation, bank, phase and transcript are consistent throughout. Its terminal-core-orbit corollary, Corollary 3.6, is also verified, with the stronger mean reciprocal of the actual fractional cover weights retained before applying the worst-case count. All covering roots lie within the computation.

3. *Uniform consequences.* The actual hop ordering, common numerical joint profile, inherited mass bounds and vanishing tails in Definition 4.1 are now formalized. Only numerical coordinates converge; no limiting geometry is asserted. Theorem 4.2 is also verified, using actual prefixes, rounded widths and the full outside-family budget. The order is fixed finite parameters, dimension limit, cutoff limit, then vanishing width error. The distinguished and protected cutoffs are independent, and the critical-moment conclusion is literal divergence of nonnegative partial sums. The uniform one-exception deficit in Corollary 4.3 is now verified as well, with constants independent of the positive entry width. A reusable sequential-exclusion lemma turns the absence of a near-full sequence into uniform constants, preserving any additional class restriction. Corollary 4.4 is verified in the stronger form requiring only finitely many leading-rank bounds with positive slack. Its power-envelope case reuses Mathlib's p-series theorem and the moment-to-tail estimate. The finite-skeleton exclusion is verified for every $q \geq 0$, using an explicit zero profile and reclassification of the same computation. Its

contrapositive counts families with more than εL hops. Theorem 4.6 is now verified, including its inverse-square probability bound. It retains the actual mean reciprocal cover through both upper limits; the earlier hop-tail inequality follows by the terminal-orbit bound. The stronger form permits any fixed legal modulus and independent cutoffs, and its probability consequence needs only an envelope on the limiting profile. Both uniform cover criteria in Corollary 4.7 are now verified, including the real-power case for every $p > 1$. A more general criterion permits any summable envelope whose cover-budget-weighted tail has lower limit zero. Finite Cauchy, Mathlib's integral comparison and the existing uniform-gap theorem supply the proof; the same chosen order witnesses both class restrictions. Do not replace the ordered limits by a single growing finite certificate.

4. *Two-exception refinements.* The complete continuation suppliers in Proposition 3.7 are now verified. Their stronger forms retain actual orbit sizes and allow interval covers with the quadratic erosion penalty. Corollary 4.8 is also verified, retaining the mean quadratic penalty through the dimension and prefix lower limits. It permits independent cutoffs and any fixed legal modulus; the original linear loss follows directly. The full shared-support payment in Proposition 5.1 is now verified. It retains the actual raw union at the recovery stage, and the finite mean-support bound uses the smaller older-bank movement constant while preserving the full residual-cut cost. The global seed set ranges over all actual output-bit times, and its raw payment is valid for every shift. The next target is the uniform global-sharing criterion in Theorem 5.2. Its dimension and prefix limits must use the same joint-profile subsequence; its good cutoffs need only be arbitrarily large. Metric criteria will then replace temporal counts with raw interval covers.

5. *Independent construction tests.* Formalize the serial native geometry as an exact finite construction, including all its return families, payments and arbitrary-seed cover lower bound. Separately prove the movement theorem and its specified query-interface applications. Finally formalize the causal-group support-to-flow argument, followed by its native metric application. The attached short-distance count for that application is already formalized, including its bridge to the nearest-attachment distance. These three branches do not serve as premises of the main uniform exclusion theorem.

A.2 Reuse and statement fidelity

The first, second, fourth, fifth and eighth papers contain relevant tape, dependency, entropy and owner-order libraries. The native short-pair count in Section 8 already imports the first paper's row and ball counting declarations, and strengthens its output-block count. The imported results are read from their original sources through the existing build adapter in the second paper; no copy is introduced. Family accounting imports the eighth paper's canonical family library and strengthens the bound by retaining the owner's two distances to the span endpoints. A shared weighted incidence identity supplies both span-length and variation accounting; the original Q2 estimates follow by bounding the overlap by two. The clipped-shadow module reuses those actual families and the shared monotone-patch library. Threshold crossing counts, Mathlib's finite-path triangle inequality and order-convex preimage lemmas give the variation bounds without a separate induction for each shadow word or packing. The proof retains exact crossing counts and handles metric-valued paths. The phase lemma reuses Mathlib's finite paths, cycle rotation and cardinality bounds. Shared interval geometry, periodic phase counting and finite-orbit modules keep the three arguments separate. The stronger cut deletes only original nonleaves and charges their actual lower supports. The complete orbit equality explicitly accounts for inactive actions. The packed-bank

proof reuses the patch bound and Mathlib's uniqueness of paths in a forest. A common set of rank residues labels every spine component injectively. Unordered endpoint pairs index the packed maps; list composition handles the leaf sandwich and exceptional families. The exact orbit proof permits the representing word of a packed slot to vary with the argument. All originally selected unpaid actual edges are covered, since the phase pays every lower of each deleted family. Paid residual recovery uses the shared causal induction and grid-payment lemmas. Cyclic freezing reuses finite-path variation and the first paper's transpose-position formula, constructs actual disjoint eroded gap arcs, and counts safe common orders. Sharp periodic-stripe discrepancy transfers those counts to actual demands. The second paper's entropy inequalities complete the deficit bound without new entropy machinery. The interval-cover lemma imports the shared library's greedy cover and packing certificate. It permits an empty target and arbitrary order-convex sets in a linear order. Its fractional bound uses Mathlib's finite-sum and order tools; no linear-programming development is needed. Minimum fractional covers use Mathlib's extreme value theorem on the finite weight cube. Canonical time components reuse path oscillation and have disjoint internal edge sets. The actual complete-orbit count spends distinguished owners and counts the common root once. Fractional weights are transferred to these shared components before interval rounding; the actual circular-gap estimate uses finite edge counting and Cauchy–Schwarz. No cover-dependent merging or new entropy lemma is needed. Anchored erosion uses arbitrary real lengths on a finite circular subdivision. The sorted gaps partition its elementary edges. Anchoring preserves the old gap relation on pairs of uncovered edges, and the shared partition-energy lemma charges exactly the pairs touching the covered union. Real-coordinate telescoping and a unit-weight specialization identify the area definitions. The formal model is this finite subdivision, not an intrinsic AddCircle or Haar-measure construction. The paid-cover assembly reuses these results and Mathlib's finite expectation identities. The subdivision-refinement adapter preserves raw coordinates and their actual gap area; boundedness, rather than attainment, justifies the supremum over chosen witnesses. Local record recovery and whole-record capacity supply the same transcript for every output. A common paid-averaging theorem accepts any valid raw-gap credits; the terminal-orbit corollary reuses it with exact fractional covers. A shared unit-weight cover lemma and regression of the actual roots supply the finite fractional upper bound without a new covering premise. For the remaining results, inspect exact types and dependency pins before choosing imports. Use sibling packages through relative paths and shared lemmas through the repository's common library; do not duplicate the earlier developments or change their frozen theorem statements.

The following distinctions are part of the specification: a physical head is not a return family; an endpoint witness is not a record with free capacity; a source overcover is not a set of independent bits; and a numerical subsequential profile is not a limiting geometry. The movement theorem quantifies over order and matching but retains its specified query interface. The causal theorem assumes local functional separation on every helper fibre, not merely separated final outputs. The serial example has a prescribed geometry and one root and phase, not an implemented transposer or an averaged counterexample.

After each formalization stage, compare the entire English statement with the Lean type, including inherited model assumptions and the order of quantifiers. The axiom commands and declaration index are generated from the actual formal citations and must be regenerated when those citations change. Neither the statement index nor a successful PDF build constitutes formal verification.

B Statement index

The manuscript contains 32 numbered results and 6 numbered definitions. Lean-verified results: 22; results with complete written proofs only: 10. A star marks a Lean-verified statement. Stable labels below identify the statements; they are not Lean declaration names. Source files and statement fingerprints are listed in `FORMALIZATION-INDEX.md`.

Reference	Statement	Stable label
Definition 2.1*	Complete families and the two-class condition	`def:complete-families`
Definition 2.2*	Admissible return geometry	`def:admissible-geometry`
Lemma 2.3*	Family accounting	`lem:family-accounting`
Definition 2.4*	Paid clipping and selected orbits	`def:clipped-shadow`
Lemma 2.5*	Variation of a shadow or successful path	`lem:shadow-variation`
Lemma 2.6*	The Q2 overlap forest and protected phases	`lem:protected-phase`
Theorem 2.7*	Theorem S: an exact source bank	`thm:exact-source-bank`
Lemma 2.8*	Paid residual-source recovery	`lem:residual-source-recovery`
Theorem 2.9*	Theorem F: cyclic freezing	`thm:cyclic-freezing`
Definition 3.1*	Fractional complete-core cover	`def:fractional-core-cover`
Lemma 3.2*	Fractional interval covers	`lem:interval-rounding`
Lemma 3.3*	A fractional raw-source gap bound	`lem:fractional-gap`
Lemma 3.4*	Anchored erosion of circular gaps	`lem:anchored-erosion`
Theorem 3.5*	Theorem C: complete covers with the full paid cost	`thm:paid-source-cover`
Corollary 3.6*	Counting terminal core orbits	`cor:orbit-count-deficit`
Proposition 3.7*	Complete continuation suppliers	`prop:continuation-covers`
Definition 4.1*	Near-full sequence and joint profile	`def:joint-profile`
Theorem 4.2*	Theorem U: necessary hop tails	`thm:necessary-hop-tail`
Corollary 4.3*	One arbitrary exceptional family	`cor:one-exception`
Corollary 4.4*	Common size envelopes	`cor:common-envelope`
Corollary 4.5*	A fixed finite skeleton	`cor:finite-skeleton`
Theorem 4.6*	Necessary reciprocal-cover bound	`thm:necessary-cover`
Corollary 4.7*	Uniform cover criteria	`cor:uniform-cover`
Corollary 4.8*	Necessary complete-observable movement	`cor:necessary-exposure`
Proposition 5.1*	The full shared-support payment	`prop:finite-sharing`
Theorem 5.2*	Theorem H: uniform exclusion by global sharing	`thm:uniform-sharing`
Corollary 5.3	Necessary global continuation density	`cor:sharing-density`
Proposition 5.4	Metric shared-support criterion	`prop:metric-sharing`
Corollary 5.5	Sublinear raw covering exponents	`cor:covering-exponents`
Proposition 6.1	Serial exceptional spans can require a quadratic cover	`prop:serial-quadratic-cover`
Theorem 7.1	Theorem M: movement to cover distinct query types	`thm:movement`
Corollary 7.2	Queries spread over a column interval	`cor:movement-interval`
Proposition 7.3	Exact helper and gap-area calibration	`prop:subset-calibration`
Corollary 7.4	The dimension limit precedes the radius limit	`cor:subset-movement`

Reference	Statement	Stable label
Definition 8.1	Causal group typing	`def:causal-group-typing`
Theorem 8.2	Theorem Q: paid routing from causal groups	`thm:causal-group-routing`
Lemma 8.3*	Attached short-distance count	`lem:attached-short-count`
Corollary 8.4	Narrow causal groups require an almost complete helper	`cor:causal-group-obstruction`

Verified declarations. The following declarations are audited by the generated `lean/PaperAxioms.lean`. They depend at most on `propext`, `Classical.choice` and `Quot.sound`. Earlier-paper declarations cited for reuse are verified there.

`SourceCovers/FamilyAccounting.lean`:
`SourceCovers.actualOwner`, `SourceCovers.ActualFamily`, `SourceCovers.HasSpanColoring`, `SourceCovers.mem_family_edges_iff`, `SourceCovers.constant_extension_isWalk`

`SourceCovers/AdmissibleGeometry.lean`:
`SourceCovers.AdmissibleGeometry`, `SourceCovers.AdmissibleGeometry.inputLength`, `SourceCovers.AdmissibleGeometry.deficit`, `SourceCovers.AdmissibleGeometry.relativeDeficit`, `SourceCovers.AdmissibleGeometry.orbit_count_deficit`

`SourceCovers/FamilyAccounting.lean`:
`SourceCovers.family_owner_inside`, `SourceCovers.family_hops_nested`, `SourceCovers.family_span_hop`, `SourceCovers.family_size_le_owner_arms`, `SourceCovers.two_mul_sum_family_size_le`, `SourceCovers.sum_family_variation_eq`, `SourceCovers.q2_accounting`, `SourceCovers.mem_boundaryLowers`, `SourceCovers.boundary_lowers_le_crossings`, `SourceCovers.boundary_lowers_le_family_sizes`, `SourceCovers.boundary_lowers_le`

`SourceCovers/ClippedShadows.lean`:
`SourceCovers.ActualFamily.shadow`, `SourceCovers.ActualFamily.shadow_at_hop`, `SourceCovers.ActualFamily.shadow_exact`, `SourceCovers.shadowOrbit`

`SourceCovers/ClippedShadows.lean`:
`SourceCovers.jointShadow_eq_on`, `SourceCovers.jointShadow_variation`, `SourceCovers.shadow_success_cons_iff`, `SourceCovers.shadow_word_structure`, `SourceCovers.shadow_word_variation`, `SourceCovers.variation_comp_le_crossings`, `SourceCovers.exists_sharp_patch`

`SourceCovers/ProtectedPhases.lean`:
`SourceCovers.q2_caterpillar`, `SourceCovers.protected_survives`, `SourceCovers.admissiblePhases_card`, `SourceCovers.phase_payment_sum`, `SourceCovers.phase_average_exact`, `SourceCovers.protected_phase_average`, `SourceCovers.nonleaf_mass_le_total`, `SourceCovers.phase_path_length`, `SourceCovers.phase_success_length`, `SourceCovers.phase_shadowOrbit_exact`, `SourceCovers.phase_orbit_chains`, `SourceCovers.phase_orbit_card`

`SourceCovers/ExactSourceBank.lean`:
`SourceCovers.exact_source_bank`, `SourceCovers.exactSourceBank_length`, `SourceCovers.exactSourceBank_bounded`, `SourceCovers.exactSourceBank_exact`, `SourceCovers.exactSourceBank_cover`, `SourceCovers.exactSourceBank_raw_root`, `SourceCovers.exactSourceBank_raw_values_card`, `SourceCovers.exactSourceBank_raw_selected_edge`, `SourceCovers.exactSourceBank_raw_variation`

`SourceCovers/ResidualRecovery.lean`:
`SourceCovers.paid_residual_recovery`

`SourceCovers/CyclicFreezing.lean`:
`SourceCovers.cyclic_freezing`, `SourceCovers.cyclic_freezing_oscillation`, `SourceCovers.cyclic_freezing_genie`, `SourceCovers.cyclic_freezing_genie_raw`

SourceCovers/CoreFractionalCover.lean:
SourceCovers.coreRawSets, SourceCovers.fractionalCoreCover,
SourceCovers.fractionalCoreCover_exists_optimal,
SourceCovers.one_le_fractionalCoreCover_toReal,
SourceCovers.coreRawSets_phase_shadowOrbit_eq,
SourceCovers.fractionalCoreCover_ne_top_iff_subset,
SourceCovers.fractionalCoreCover_exactSourceBank_ne_top
SourceCovers/IntervalCover.lean:
SourceCovers.exists_cover_packing, SourceCovers.exists_subcover_le_weight
SourceCovers/FractionalGap.lean:
SourceCovers.fractional_phase_gap, SourceCovers.fractional_core_gap,
SourceCovers.fractional_core_gap_of_union_variation,
SourceCovers.fractional_core_gap_of_weights
SourceCovers/AnchoredErosion.lean:
SourceCovers.penalty, SourceCovers.area_lower_bound, SourceCovers.area_lower_bound_linear,
SourceCovers.anchored_displacement_area_lower_bound,
SourceCovers.anchored_point_area_lower_bound
SourceCovers/PaidSourceCover.lean:
SourceCovers.complete_cover_paid_deficit, SourceCovers.paid_deficit_of_gap_credit,
SourceCovers.phaseCoverCredit_bounds, SourceCovers.meanCoverCredit
SourceCovers/OrbitCountDeficit.lean:
SourceCovers.phaseCoreCover, SourceCovers.meanReciprocalCover,
SourceCovers.meanReciprocalCover_bounds, SourceCovers.reciprocal_cover_paid_deficit,
SourceCovers.orbit_count_paid_deficit
SourceCovers/ContinuationClusters.lean:
SourceCovers.oneExceptionBank_exact, SourceCovers.twoExceptionBank_exact,
SourceCovers.continuationSuppliers, SourceCovers.continuationSuppliers_exact,
SourceCovers.continuationSuppliers_exact_bank, SourceCovers.continuationSuppliers_root,
SourceCovers.continuationSuppliers_past, SourceCovers.olderLandingSuppliers,
SourceCovers.olderLandingSuppliers_exact,
SourceCovers.continuationSuppliers_fractional_bound,
SourceCovers.olderLandingSuppliers_fractional_bound,
SourceCovers.continuation_fractional_bounds, SourceCovers.clusterVariation,
SourceCovers.continuationClusters_erosion, SourceCovers.continuationClusters_gap,
SourceCovers.continuationClusters_gap_linear
SourceCovers/JointProfiles.lean:
SourceCovers.NearFullSequence, SourceCovers.NearFullSequence.ofGeometry,
SourceCovers.NearFullSequence.JointProfile,
SourceCovers.NearFullSequence.exists_jointProfile,
SourceCovers.NearFullSequence.JointProfile.hop_mass,
SourceCovers.NearFullSequence.JointProfile.variation_mass,
SourceCovers.NearFullSequence.JointProfile.hop_antitone,
SourceCovers.NearFullSequence.JointProfile.hop_le_rank,
SourceCovers.NearFullSequence.JointProfile.finite_sums_tendsto,
SourceCovers.NearFullSequence.JointProfile.prefix_add_tail,
SourceCovers.NearFullSequence.JointProfile.finite_tails_tendsto,
SourceCovers.NearFullSequence.JointProfile.tails_tendsto_zero
SourceCovers/NecessaryHopTails.lean:
SourceCovers.NearFullSequence.JointProfile.necessary_hop_tail,
SourceCovers.NearFullSequence.JointProfile.tail_floor,
SourceCovers.NearFullSequence.JointProfile.not_summable_moment,
SourceCovers.NearFullSequence.JointProfile.moment_partial_sums_tendsto_atTop

SourceCovers/UniformDeficit.lean:
SourceCovers.one_exception_uniform_deficit, SourceCovers.uniform_deficit_of_no_nearFull,
SourceCovers.NearFullSequence.one_exception_false,
SourceCovers.AdmissibleGeometry.deficit_card_iff

SourceCovers/CommonEnvelopes.lean:
SourceCovers.EnvelopeTailSmall, SourceCovers.finite_prefix_envelope_uniform_deficit,
SourceCovers.common_envelope_uniform_deficit, SourceCovers.power_envelope_tailSmall,
SourceCovers.power_envelope_uniform_deficit

SourceCovers/FiniteSkeleton.lean:
SourceCovers.AdmissibleGeometry.withSkeleton,
SourceCovers.NearFullSequence.empty_core_false, SourceCovers.empty_core_uniform_deficit,
SourceCovers.finite_skeleton_uniform_deficit,
SourceCovers.large_family_count_of_small_deficit

SourceCovers/NecessaryCover.lean:
SourceCovers.NearFullSequence.JointProfile.reciprocalCoverLimit,
SourceCovers.NearFullSequence.JointProfile.reciprocalCoverLimit_bounds,
SourceCovers.NearFullSequence.JointProfile.weighted_reciprocal_cover,
SourceCovers.NearFullSequence.JointProfile.necessary_reciprocal_cover,
SourceCovers.NearFullSequence.JointProfile.necessary_reciprocal_cover_default,
SourceCovers.NearFullSequence.JointProfile.coverProbabilityLimit,
SourceCovers.NearFullSequence.JointProfile.coverProbabilityLimit_le_reciprocal,
SourceCovers.NearFullSequence.JointProfile.hop_le_of_pointwise,
SourceCovers.NearFullSequence.JointProfile.inverse_square_cover_probability

SourceCovers/UniformCover.lean:
SourceCovers.AdmissibleGeometry.HasMeanCoverBound, SourceCovers.CoverTailSmall,
SourceCovers.uniform_deficit_of_no_ordered_nearFull,
SourceCovers.linear_mean_cover_uniform_deficit,
SourceCovers.envelope_mean_cover_uniform_deficit, SourceCovers.power_cover_tailSmall,
SourceCovers.power_mean_cover_uniform_deficit

SourceCovers/NecessaryMovement.lean:
SourceCovers.phaseMovement, SourceCovers.NearFullSequence.JointProfile.movementLimit,
SourceCovers.NearFullSequence.JointProfile.completeMovement_bounds,
SourceCovers.NearFullSequence.JointProfile.necessary_movement_penalty,
SourceCovers.NearFullSequence.JointProfile.movementPenalty_le_complete,
SourceCovers.NearFullSequence.JointProfile.necessary_complete_movement_default,
SourceCovers.NearFullSequence.JointProfile.scaled_complete_movement_liminf

SourceCovers/GlobalSharing.lean:
SourceCovers.shared_support_count_deficit, SourceCovers.shared_support_paid_deficit,
SourceCovers.phase_shared_deficit, SourceCovers.sharedRawSupport,
SourceCovers.sharedSeedCount, SourceCovers.exactSourceBank_shared_cover,
SourceCovers.sharedRawSupport_card_le_sum, SourceCovers.sharedRawSupport_blocks_card_le,
SourceCovers.exists_paidBank_shared_deficit

SourceCovers/UniformSharing.lean:
SourceCovers.sharing_uniform_deficit, SourceCovers.liminf_sharing_uniform_deficit,
SourceCovers.sublinear_sharing_uniform_deficit, SourceCovers.power_sharing_uniform_deficit,
SourceCovers.AdmissibleGeometry.HasSharingBounds,
SourceCovers.NearFullSequence.JointProfile.prefix_sharing_limit,
SourceCovers.NearFullSequence.JointProfile.sharing_tail_lower

SourceCovers/NativeShortCount.lean:
SourceCovers.attachedShortBits, SourceCovers.card_attachedShortBits_le_local,
SourceCovers.card_attachedShortBits_le_profile, SourceCovers.card_attachedShortBits_le,
SourceCovers.attachedShortBits_zero_radius, SourceCovers.attachedShortBits_zero_block

C Building and checking this draft

The paper sources and PDF use CC BY 4.0; the Lean library uses Apache 2.0. Definitions 2.1, 2.2, 2.4, 3.1 and 4.1, Lemmas 2.3, 2.5, 2.6, 2.8, 3.2 to 3.4 and 8.3, Theorems 2.7, 2.9, 3.5, 4.2, 4.6 and 5.2, Corollaries 3.6, 4.3 to 4.5, 4.7 and 4.8, and Propositions 3.7 and 5.1 are Lean-verified. Their cited declarations depend only on `propext`, `Classical.choice` and `Quot.sound`. The remaining numbered results have complete written proofs and are not formalized. No additional axiom, admitted proof or native decision procedure is used in the new library.

The package pins Lean and Mathlib to the repository's existing v4.34.1 versions. It requires the existing second-paper build adapter, whose source directory points to the first paper's canonical Lean files. It also requires the eighth paper's `boundedNesting` package for actual families and the shared `lean-machine/` package for interval covers, finite incidence sums, monotone patches, thin interval graphs, protected phases, finite action orbits, uniform source banks, shifted-grid payment, causal recovery, circular gaps, periodic stripes, minimum fractional covers, canonical time components, weighted circular subdivisions, anchored partition-energy bounds, profile compactness, finite-error normalization, weighted-tail comparison and uniform gaps from sequential exclusion, extended lower limits, power-envelope moments, bounded upper limits, inverse-square tails and quantitative real-power tails. The joint-profile proofs reuse Mathlib's compact products, finite-sum limits, summability, normalized ceilings and nonnegative-series divergence. The uniform-deficit proof reuses reciprocal limits, the squeeze theorem and the logarithm/real-power equivalence. Common envelopes add Mathlib's extended-nonnegative lower limits and p-series theorem. The supplied family ordering is retained directly through the sequence construction. The finite-skeleton proof instead supplies the zero profile explicitly; its family-count consequence reuses Mathlib's natural-floor inequalities. The reciprocal-cover proof retains actual averages through both limits and reuses Mathlib's subsequence characterization of an upper limit. The probability bound uses a finite indicator comparison and the existing finite inverse-square sum estimate. The uniform mean-cover criteria reuse Mathlib's finite Cauchy inequality, integral comparison and improper-integral formula, followed by the existing uniform-gap adapter. A chosen family order is retained for both the envelope and the cover budget. The continuation suppliers reuse ordered-bank exactness, finite maximal elements obtained by greatest time, fractional-cover feasibility, interval oscillation and anchored erosion. Their source-bank slots include complete older continuations, and missing exceptions are identity maps. The complete-movement proof reuses finite interval hulls and the existing paid-error limit, now shared with the reciprocal-cover argument. Explicit bank-variation bounds justify the two real lower limits. The final radius lower limit uses the extended nonnegative reals to allow divergence. The shared-support proof reuses complete-closure maximality, residual-shift selection, joint record/raw entropy and signed phase averaging. A finite union of clipped grid blocks is paid once, with a bound valid for every shift; the older bank's movement never replaces the full bank's residual bill. Finite entropy is reused from the second paper's library. The uniform-sharing proof retains the same original-order prefixes and uses the generalized paid-error limit. A shared quadratic tradeoff handles zero cost coefficients and arbitrary common good cutoffs. Mathlib supplies extended-real scaling and the decay of positive real powers. Readers need the whole repository, with sibling folders in place. No source from an earlier paper is copied or modified. From `lean/`, run

```
lake exe cache get
lake build
lake env lean PaperAxioms.lean
```

The cache command is needed only when the pinned dependency objects are not already available. The axiom output must contain no axioms beyond the three listed above. No heartbeat override is

needed for the new proofs.

From this paper's top-level directory, regenerate or check the statement index and axiom commands with

```
python3 tools/statement_index.py
python3 tools/statement_index.py --check
```

The script reads the section order from `paper/main.tex`, checks stable labels, proof-status markers and the corresponding proof or outline environments, and generates the index appendix, `FORMALIZATION-INDEX.md`, and `lean/PaperAxioms.lean`. It does not itself verify the mathematics or execute the axiom commands. `FORMALIZATION.md` records completed work and the remaining plan.

Build the PDF from `paper/` with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

Resolve undefined references and citations, and inspect the rendered pages. Earlier-paper references are pinned in `paper/references.bib`, and their proof status is stated where used. Public paths are relative to this repository. Run `tools/check_public_paths.py` from the repository root as described in `CONTRIBUTING.md` before publishing. That scan checks portability and local-path disclosure, not theorem correctness.

References

[1] Hans Brende and Claude Opus 5.5 (Anthropic). Bounded nesting and depth reduction for matrix transposition on tapes. Manuscript and accompanying Lean formalizations, repository HansBrende/papers, October 2026. Pinned revision 5cf90b9e346fee3bb5d7fa233f5b1f1dfec5f447; proof status is specified result by result. URL: https://github.com/HansBrende/papers/tree/5cf90b9e346fee3bb5d7fa233f5b1f1dfec5f447/Bounded-Nesting-and-Depth-Reduction-for-Matrix-Transposition-on-Tapes.

[2] Hans Brende and Claude Opus 5.5 (Anthropic). Curved motion, depth reduction, and storage codes for matrix transposition on tapes. Manuscript and accompanying Lean formalizations, repository HansBrende/papers, October 2026. Pinned revision 5cf90b9e346fee3bb5d7fa233f5b1f1dfec5f447; proof status is specified result by result. URL: https://github.com/HansBrende/papers/tree/5cf90b9e346fee3bb5d7fa233f5b1f1dfec5f447/Curved-Motion-Depth-Reduction-and-Storage-Codes-for-Matrix-Transposition-on-Tapes.

[3] Hans Brende and Claude Opus 5.5 (Anthropic). Matrix transposition and the tape coding conjecture. Manuscript and accompanying Lean formalizations, repository HansBrende/papers, October 2026. Pinned revision 5cf90b9e346fee3bb5d7fa233f5b1f1dfec5f447; proof status is specified result by result. URL: https://github.com/HansBrende/papers/tree/5cf90b9e346fee3bb5d7fa233f5b1f1dfec5f447/Matrix-Transposition-and-the-Tape-Coding-Conjecture.

[4] Hans Brende and Claude Opus 5.5 (Anthropic). Nested elimination and half-rate bounds for storage codes. Manuscript and accompanying Lean formalizations, repository HansBrende/papers, October 2026. Pinned revision 5cf90b9e346fee3bb5d7fa233f5b1f1dfec5f447; proof status is specified result by result. URL: https://github.com/HansBrende/papers/tree/5cf90b9e346fee3bb5d7fa233f5b1f1dfec5f447/Nested-Elimination-and-Half-Rate-Bounds-for-Storage-Codes.

[5] Hans Brende and GPT-6 Astra (OpenAI). Directed recovery structures in tape computations. Manuscript and accompanying Lean formalizations, repository HansBrende/papers, October 2026. Pinned revision 5cf90b9e346fee3bb5d7fa233f5b1f1dfec5f447; proof status is specified result by result. URL: https://github.com/HansBrende/papers/tree/5cf90b9e346fee3bb5d7fa233f5b1f1dfec5f447/Directed-Recovery-Structures-in-Tape-Computations.

[6] Hans Brende and GPT-6 Astra (OpenAI). Local coding barriers and the limits of tape geometry for matrix transposition. Manuscript and accompanying Lean formalizations, repository HansBrende/papers, October 2026. Pinned revision 5cf90b9e346fee3bb5d7fa233f5b1f1dfec5f447; proof status is specified result by result. URL: https://github.com/HansBrende/papers/tree/5cf90b9e346fee3bb5d7fa233f5b1f1dfec5f447/Local-Coding-Barriers-and-the-Limits-of-Tape-Geometry-for-Matrix-Transposition.

[7] L. R. Ford, Jr. and D. R. Fulkerson. Maximal flow through a network. *Canadian Journal of Mathematics*, 8:399–404, 1956. doi:10.4153/CJM-1956-045-5.

[8] David Harvey and Joris van der Hoeven. Integer multiplication is at least as hard as matrix transposition. arXiv:2503.22848, 2025. URL: https://arxiv.org/abs/2503.22848.

[9] Qifa Yan, Minquan Cheng, Xiaohu Tang, and Qingchun Chen. On the placement delivery array design in centralized coded caching scheme. *IEEE Transactions on Information Theory*, 63(9):5821–5833, 2017. URL: https://arxiv.org/abs/1510.05064v3, doi:10.1109/TIT.2017.2725272.